

Exploring and enhancing the volumetric behaviour in solar receivers through specular reflectivity and simplified design

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ABSTRACT

One of the remaining challenges of point focusing concentrating solar power systems is the realization of a true volumetric receiver, one whose entire volume is utilized for the absorption of irradiance. Commercial state-of-the-art receivers (e.g., HiTRec-II and SolAir-200) have not demonstrated the volumetric effect, because of low radiation penetration within the absorber. Then, literature state-of-the-art receivers were able to demonstrate a volumetric behaviour (toward achieving the volumetric effect) through axially varied macro-scale porosity (void fraction), but at the cost of design complexity and manufacturability. This study presents the conceptual design and computational assessment of an improved volumetric receiver by employing varied reflectivity on the irradiated surfaces, thereby enhancing radiation penetration. This approach maintains the simplicity, manufacturability, and commercial viability of structured receivers. Three reflectivity distributions were initially investigated using a 3D heat transfer mathematical model, verified with literature experimental data, which incorporated a source term ascertained through Monte-Carlo Ray Tracing (MCRT). The findings from these three design cases revealed that while diffuse reflectivity distributions could improve thermal efficiency, they were unable to enhance radiation penetration. Consequently, an improved volumetric solar receiver was realized by integrating specular reflectivity, leading to a more uniform source term distribution and achieving a thermal efficiency of 96%, as opposed to the 85% thermal efficiency of the commercial state-of-the-art HiTRec-II receiver. The discovery of the specular receiver as a true volumetric receiver facilitated an exploration of quantitative definitions of the volumetric effect. The exploration concluded that volumetric effect metrics would only target designs that maximizing thermal efficiency when the receiver exhibits a truly volumetric nature, characterized by a uniform source term distribution.

1. Introduction

To face the persistent challenges of global warming and rising global energy demands, research efforts in the past few decades have focused on routes to utilizing different sources of renewable energy. One route that proved to be promising in large-scale tests is Concentrated Solar Power (CSP). A CSP plant consists of a system of reflectors that focus radiation onto a solid receiver, where radiative energy is absorbed by the solid and transferred by convection to a working fluid (heat transfer fluid – HTF) for either thermal storage or conversion into electricity. CSP's compatibility with cheap thermal storage, hybridization, and well-developed thermal-to-electrical thermodynamic cycles highlight the technology's significance and potential over other renewable technologies [1,2].

Based on the concentrator configuration, CSP systems can be broadly

categorized into line-focusing (or tube-receiver) systems and point-focusing (or central receiver) systems. Point-focusing CSP systems are of more interest, because their higher concentration ratios can achieve higher HTF temperatures and in turn greater solar-to-electric conversion efficiencies [3,4]. Additionally, because the thermal-to-electric conversion step is governed by the principles of operation for any power plant, the solar-to-thermal conversion step becomes more of a bottleneck. Subsequently, the design and optimization of the receiver system – where the solar-to-thermal energy conversion occurs – is a key research area in central CSP [5].

Volumetric receivers, commonly used in central CSP systems (indicative examples in Table 1), are porous structures that are meant to absorb concentrated irradiance with their entire interior volume and transfer that heat to the HTF passing through them. However, state of the art of volumetric receivers like HiTRec-II and SolAir-200 did not exhibit full utilization of the receiver volume. Ninety percent of the

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Nomenclature:*Greek Symbols*

$\bar{\beta} = -1/\rho(\partial\rho/\partial T) _p$	thermal expansion coefficient evaluated at $\bar{T}$ [1/K]
ε	surface emissivity [-]
μ	dynamic viscosity [Pa.s]
$\bar{\rho}_g$	fluid density at the average wall temperature $\bar{T}$ [kg/m ³]
σ	work done per unit volume (i.e. molecular stress) [J/m ³]
σ_B	Stefan-Boltzmann constant [W/m ² .K ⁴]
ϕ_s	porosity of the solid material [-]
ϕ	bulk porosity or void fraction of the receiver [-]

Latin Symbols

C_p	thermal heat capacity [J/kg.K]
L_{ch}	Length of channel or absorber module [mm]
P	Radiative power (per receiver module unless otherwise stated) [W]
Q	Absolute heat flow or radiative power (per receiver module unless otherwise stated) [W]
R_{ch}	Radius of a cylinder inscribed in a non-cylindrical channel (equivalent to half the width of a square channel) [mm]
$\bar{T}$	average temperature of walls considered in the direction of buoyant force [K]
$n_{prim,ab} _i$	number of absorbed primary rays by surface element 'i' [-]
ρ_d	diffuse reflectivity
q	diffusive heat flux (i.e. conductive flux) [W/m ²]
$\mathbf{u}$	velocity vector of translational motion [m/s]
I	incident radiation flux (irradiation) [W/m ²]
K	kinetic energy per unit mass [J/kg]
S	additional volumetric energy sources [W/m ³]
e	internal energy per unit mass [J/kg]
F	the volume or bulk force vector [N/m ³]

Subscripts

g	gas domain
s	solid domain
i	pertaining to surface element 'i'
m	mutual (coming from other surfaces in the geometry)
amb	ambient
ext	external (coming from external sources)
$prim$	primary sources
ray	per ray
$lost/loss$	lost to ambient environment
rad	radiative or secondary sources

Abbreviations

BHS	Boundary Heat Source
CFD	Computational Fluid Dynamics
CSP	Concentrating Solar Power
DLR	German Aerospace Institute
HFSS	High Flux Solar Simulator
HiTRec-II	High Temperature Receiver – II
HTC	Heat Transfer Coefficient
HTF	Heat Transfer Fluid
IR	Infra-red
LMTD	Log Mean Temperature Difference
MCRT	Monte Carlo Ray Tracing
D	Demonstrated performance
S	Simulated performance
NMB	Normalized Mean Bias
NMSD	Normalized Mean Standard Deviation
OAVR	Open Air Volumetric Receiver
PPI	Pores Per Inch
RMSE	Root Mean Square of Errors

radiation was absorbed in the front 20% of the receiver for an absorber modelled by Alberti, *et al.* [6] based on the dimensions of the SolAir-200. Full utilization of receiver volume is characterized by the “volumetric effect”, which is qualitatively described as the solid receiver temperature at the inlet being lower than the gas exit temperature. It is expected to occur due to a high rate of convective cooling at the inlet, and an – ideally – uniformly distributed radiative source term. The volumetric effect is highly sought after, as a lower frontal solid temperature would minimize re-radiation losses and improve thermal efficiency. There are no established quantitative definitions for the volumetric effect to the best of the authors’ knowledge.

In an effort to realize the volumetric effect, numerical studies have looked into the influence of geometric and optical parameters (like macro-scale porosity or void fraction, pore size and spectral selectivity) on the temperature profiles and efficiency – i.e. the volumetric behaviour - of isotropic volumetric absorbers like foam, honeycomb and wire mesh receivers. Many numerical studies have been performed on ceramic foam receivers. Though most agree that a higher porosity is favourable, as it improves radiation penetration, there is no consensus on the effect of pore size on thermal efficiency or volumetric behaviour. Experimental studies on foam receivers have reached no consensus, either [7–9]. The same can be said for wire mesh receivers [10], and honeycomb receivers in numerical [11–13] and experimental studies [13,14]. Also, despite the numerical predictions of Wu, *et al.* [15], none of these isotropic absorbers have demonstrated the volumetric effect in experiments. This encouraged the exploration of novel non-isotropic absorbers.

In this vein, some works have focused on enhancing the convective

heat transfer coefficient of the receiver to attain volumetric behavior – considering that convection is practically the only mechanism facilitating heat transfer to the fluid phase. Gomez-Garcia, *et al.* [16] designed a set of honeycomb receivers with cutback features at the inlet to reduce pressure drop, and Fend, *et al.* [17] considered an absorber cup with shorter channels at the center for the same reason. Both observed a reduced pressure drop and marginal improvements in thermal efficiency, but at the cost of lowering the inlet heat transfer coefficient (HTC) and, in-turn, increasing the frontal solid temperature. Others have reported more success, when placing convection or turbulence enhancing features near the front of the receiver – where the maximum temperature driving force exists [7,18]. These latter works, however, were experimental and did not report axial temperature profiles. Hence, the presence of the volumetric effect could not be investigated.

Another group of works have considered axially graded-porosity designs, where the porosity of the absorber is higher at the inlet to allow for deeper radiation penetration and lower towards the back to create a denser region with a higher HTC. Chen, *et al.* [19] attempted stacking a 2-layer ceramic foam, with Zaversky, *et al.* [20] adding a third layer of stacked ceramic foam, while Avila-Marin, *et al.* [21] also incorporated three layers but of wire meshes to vary porosity. Du, *et al.* [22] modeled and optimized a gradual porosity foam receiver. Most of these works observed that a very large and sudden drop in porosity at around 60% of the receiver’s length is the most favorable configuration. Meanwhile, Zaversky, *et al.* [20] did not notice any advantage to varying porosity, possibly due to the limited ranges selected. Avila-Marin, *et al.* [23] tested a number of linearly graded porosity receivers with different repeating structures, but the only one that exceeded the base case

Table 1
Indicative literature toward achieving the volumetric effect.

Receiver Material	Geometry	Specific Surface Area (m ² /m ³)	Porosity	Extinction Coefficient (m ⁻¹)	Performance*	Reference**
–	Multi-layer staggered square grid absorber	–	–	–	Reflective losses increased by 3%	[16]
–	Honeycomb with cutback features in the front face	–	–	–	η_{th} not reported	[36]
Ceramic (SiC)	Honeycomb with frustum in the center	2027	0.85	2300	$\eta_{th} = 85\%$ (D)	[17]
Ceramic (SSiC)	Modified honeycomb with orifice restrictions in each channel	2390	~ 0.76	–	$\eta_{th} = 89.3\%$ (D)	[7]
Ceramic (SSiC)	2-Layer foam with denser front	2530	20 PPI – 80 PPI	–	$\eta_{th} = 91.7\%$ (S)	[7]
Metallic (iron-based alloy 1.4767)	Exhaust engine reactor with cut wings and perforations in each channel	> 6600	> 0.90	–	$\eta_{th} = 82\%$ $T_{g,out} = 520$ °C (D)	[7]
Ceramic (SiC)	Stacked foams with gradual porosity and pore size	–	0.90 – 0.70	–	$\eta_{th} = 92\%$ $T_{g,out} = 640$ °C (D)	[7]
Ceramic (SSiC)	Dual layer foam receiver ($L_A : L_B = 10$ mm:50 mm)	–	0.93 – 0.60	–	$\eta_{th} = 86\%$ $T_{g,out} = 700$ °C (D – at $P/\dot{m} \approx 180$ [kJ/kg])	[18]
Metallic (SS 601)	Mesheres of different sizes stacked on each other	> 4500	0.84 – 0.66	–	$T_{g,out} = 990$ K (S – at $P/\dot{m} \approx 8$ [kJ/kg])	[19]
Ceramic (SiC)	Foam – axially graded porosity distribution	–	0.95 – 0.65	802.2	$\eta_{th} = 90\%$ $T_{g,out} = -$ (S – at $P/\dot{m} \approx 1080$ [kJ/kg])	[20]
Ceramic (SiC)	Linearly graded porosity Voronoi mesh	320 – 800	0.92 – 0.84	23 – 137	$\eta_{th} = 88\%$ $T_{g,out} = 660$ K (D)	[21]
Metallic (Inconel 718)	Foam – radially graded porosity distribution	510	0.80 (avg)	502.2	Improved radiative penetration	[22]
–	Foam – radially & axially graded porosity distribution	–	0.85 – 0.95	–	$\eta_{th} = 86\%$ $T_{g,out} = 530$ K (D – at $P/\dot{m} \approx 980$ [kJ/kg])	[23]
Metallic (AISI 316L SS)	Fractal-like – repeating 2D elements on 3D planes	420 – 950	0.97 – 0.48	–	$\eta_{th}^{max} = 87\%$ $T_{g,out} = 530$ K (D)	[24]
Metallic (Ti ₆ Al ₄ V)	Modified honeycomb with pin-shaped front	318 – 870 (D) 990 – 2860 (S)	0.97 – 0.71	–	$\eta_{th}^{max} = 82\%$ $T_{g,out} = 1120$ K (S – at $P/\dot{m} \approx 890$ [kJ/kg])	[25]
Ceramic (SiC and SiO ₂)	2-slab receiver with silica honeycomb backed by SiC packed bed	–	–	–	$\eta_{th} = 64\%$ $T_{g,out} = 480$ K (D)	[30]
Ceramic (partially coated alumina)	Honeycomb – square channel partially coated w/ Pyromak 2500	–	–	–	$\eta_{th} = 64\%$ $T_{g,out} = 546$ K (D)	[31]
					$\eta_{th} = 83\%$ $T_{g,out} = 860$ K (S)	
					$\eta_{th}^{max} = 89\%$ $T_{g,out} = 965$ K (D – at $P/\dot{m} \approx 810$ [kJ/kg])	[32]
					79% (S – at unknown conditions)	[34]

* Thermal efficiencies are reported for a power on aperture to mass flowrate ratio ($P/\dot{m}$) of 700 [kJ/kg], unless otherwise stated. Legend: (S) – simulated and (D) – Demonstrated.

** The performance of the receivers mentioned in [26–29] is not reported as they are classified as modified tube receivers, not volumetric receivers.

performance (honeycomb and foam receivers) was based on randomized Voronoi cells. Indicating that the internal structure of the receiver is significant – not only the macroscopic porosity distribution.

In a similar class of attempts, [24] optimized a radially varied porosity receiver that is more open in the center to accommodate the gaussian flux profile at the aperture of the receiver. And [25] considered combining this approach with the axially graded porosity to enhance radiation penetration and reduce flow resistance. Both reported improved thermal efficiencies but no volumetric behavior.

Additionally, a group of axially graded porosity designs that has been particularly successful is the so-called “fractal-like” structures, which are inspired by the porcupine [26] and modified tube receivers that add fin-like structures to the irradiated surface to increase light-trapping (examples include [27–29]. Alberti, *et al.* [6] and Luque, *et al.* [30] optically optimized and thermally tested a receiver constructed from planar triangular pieces arranged on different 3D planes to produce layers of varying porosity. Capuano, *et al.* [31] demonstrated a honeycomb-like structure that gradually phases the porosity from 0.97

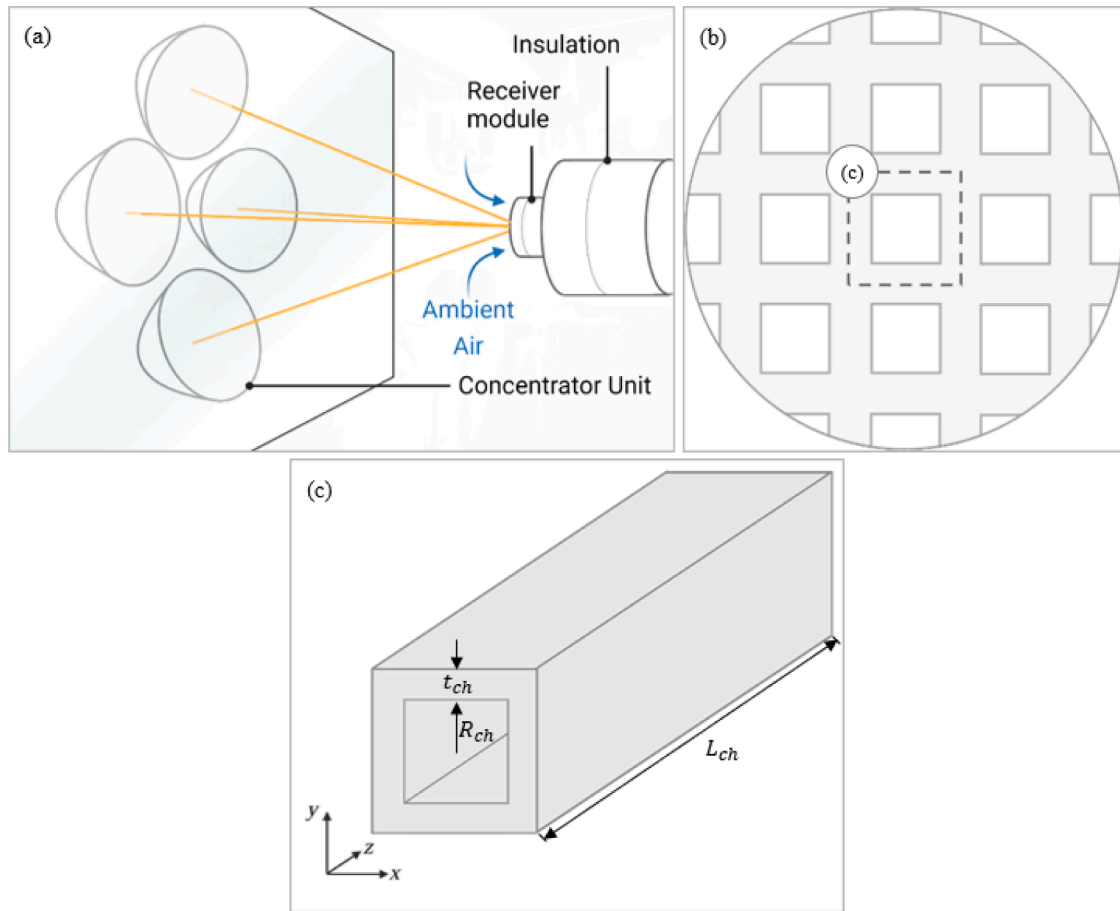


Fig. 1. System geometry. a) irradiance setup used – four lamp blocks from the DLR HFSS, b) frontal view of the square-channel honeycomb receiver geometry and c) the reduced geometry modelled in this work.

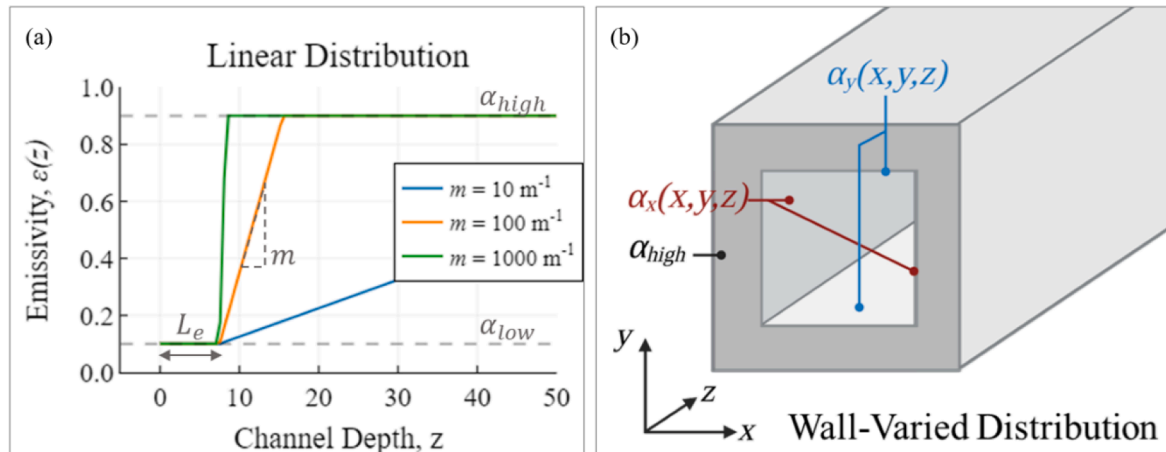


Fig. 2. Considered reflectivity distributions a) Linear axial distribution. b) Wall-varied distribution.

in a front pin-shaped section to 0.71 in a backing honeycomb section. Both the designs mentioned successfully demonstrated the volumetric effect. However, the complex designs developed required the use of layered manufacturing techniques only compatible with metals, prohibiting the use of the high-performance ceramics desired in central CSP. The effect of this materials challenge is prominent when comparing the performance of these designs to that of other novel designs presented in Table 1.

To avoid the structural complexity of varied-porosity receiver

designs, some works have focused on improving radiation penetration by modifying the optical properties of the receiver along its depth. Menigault, *et al.* [32] and Sedighi, *et al.* [33] considered packed bed absorbers with semitransparent glass beads at the front and opaque SiC particles/beads in the back. The glass beads transmit primary visible irradiance to the absorbing SiC particles and absorb secondary IR irradiance limiting re-radiation losses. Despite improved performance, these designs do not eliminate the complexity of a closed system design absorber and do not report achieving the volumetric effect. Apart from

Table 2
Variable levels used for each receiver optimization study.

Receivers	Variables	Variable Levels	
		Stage 1 Optical Param.	Stage 2 Heat Transfer Param.
Base Case	α	[0.1, 0.3, 0.5, 0.7, 0.9]	0.8
	R_{ch} [mm]	[0.5, 1, 2, 4, 8]	1
	L_{ch}/D	25	[10, 20, 40, 80]
	L_e/L_{ch}	[0.001, 0.3, 0.5, 0.7, 0.9]	0.15
Linear Distribution Case	R_{ch} [mm]	[0.5, 1, 2, 4, 8]	0.5
	m [m ⁻¹]	1000	[10, 100, 1000]
	L_{ch}/D	25	[10, 20, 40, 80]
	α_x	[0.1, 0.3, 0.5, 0.7, 0.9]	0.7
Wall-Varied Distribution Case	α_y	0.9	0.9
	R_{ch} [mm]	[0.5, 1, 2, 4, 8]	0.5
	L_{ch}/D	25	[10, 20, 40, 80]

closed systems, this approach has only been considered once in literature for open air volumetric receivers (OAVR), where Ali, *et al.* [34] modelled an alumina honeycomb receiver and applied a high absorbance coating to the back half of each channel. Though improved volumetric behavior is reported, the achieved thermal efficiency is not comparable to graded-porosity designs and the concept was not fully explored. This shows the approach has some potential to achieve the volumetric effect with the appropriate optimization procedure.

In the aim to advance renewable energy technology, an exploration and optimization of a honeycomb receiver with a novel feature of varied-reflectivity is being proposed. The goal is to enhance radiation penetration, volumetric behavior, and thermal efficiency, while

preserving the simplicity and robustness found in commercially proven receivers, such as the HiTRec-II. The absorber is to be modelled as a single channel, an approach commonly employed in structured receiver modeling [35], allowing for a detailed treatment of the source term using Monte-Carlo ray tracing (MCRT). Alongside this, a 3D steady-state mathematical model will be developed for thermal performance evaluation, with its veracity confirmed against existing literature data. A comprehensive parametric study will be conducted to determine the ideal dimensions for the receiver and to analyze the components of its energy balance, expected to guide the design towards an enhanced receiver integrating specular reflectivity. Moreover, through this work, potential quantitative definitions for the elusive volumetric effect, which currently lacks a universally accepted definition, will be explored, aiming to bring clarity to the field.

2. Methodology

This section details a description of the physical system, including its geometry and the significant phenomena involved. This is followed by the development of the mathematical model and the procedure for its verification – including the grid independence study. Finally, the parametric study designed to explore the varied-reflectivity receiver concept is described.

2.1. System description & geometry

The system used in this work consists of a high flux solar simulator (HFSS) as the radiation source, and an open absorber module that receives air and concentrated irradiance through the front. The receiver

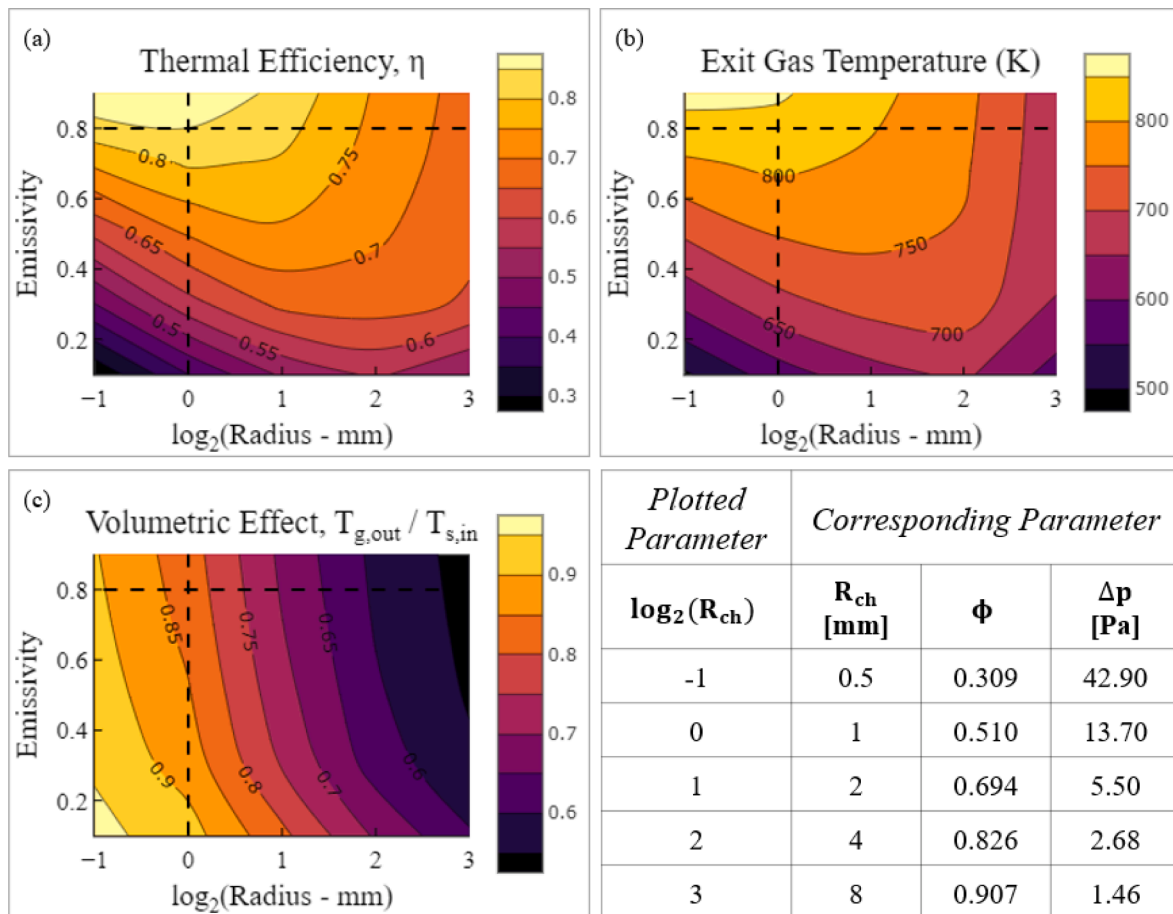


Fig. 3. Base case (Table 2) receiver optimization – Stage 1: Effect of emissivity and porosity (dashed black lines indicate the selected receiver design). Shown are contours for a) thermal efficiency, b) exit gas temperature, c) E_{vol} ratio.

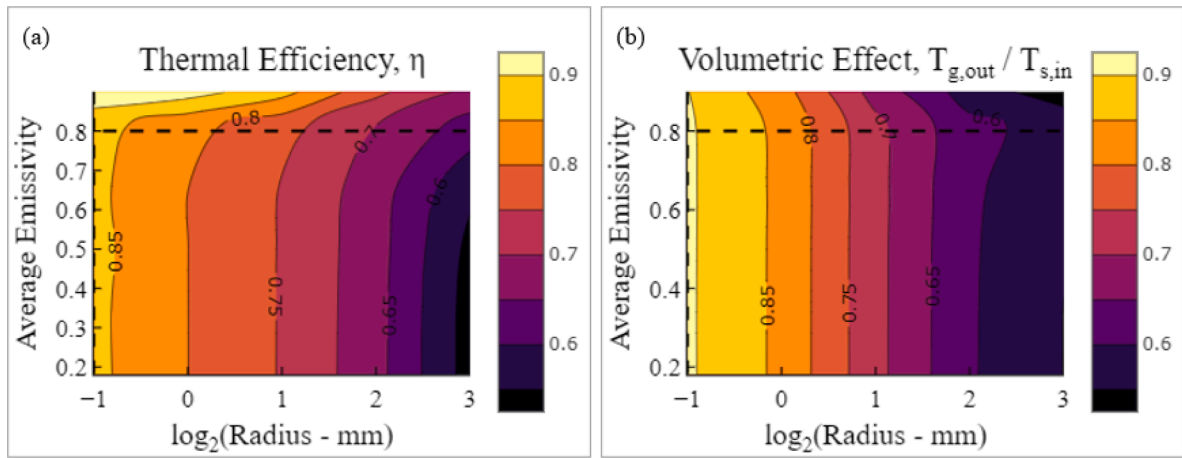


Fig. 4. Linear distribution case (Table 2) optimization – Stage 1: Effect of entry length or average emissivity and porosity (dashed black lines indicate the selected emissivity levels). Shown are contours for a) thermal efficiency, b) E_{vol} ratio.

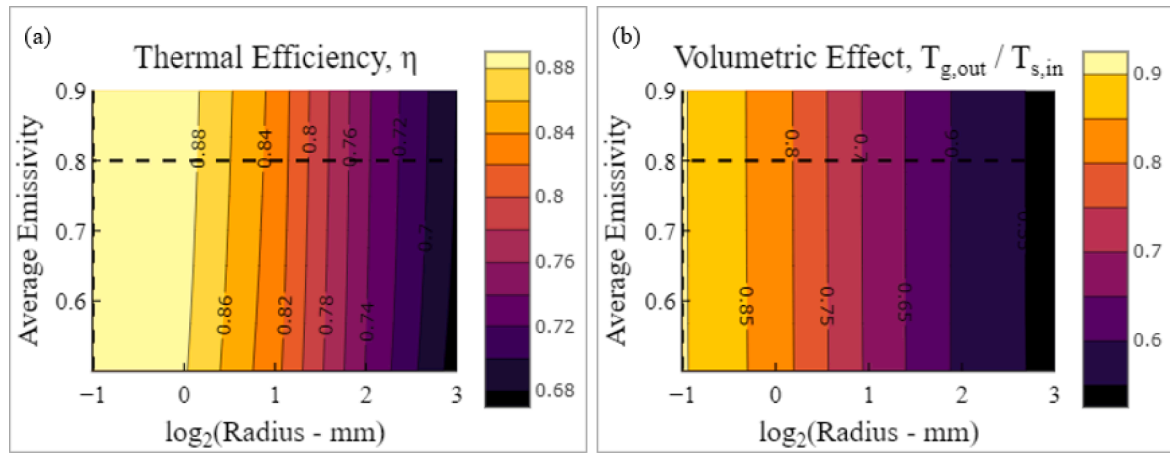


Fig. 5. Wall-varied distribution case (Table 2) optimization – Stage 1: Effect of entry length or average emissivity and porosity (dashed black lines indicate the selected emissivity levels). Shown are contours for a) thermal efficiency, b) E_{vol} ratio.

considered in this study is a square channel honeycomb receiver as those have low pressure drop, good specific surface area, and allows comparison to the state of the art (HiTRec-II) – see Fig. 1b. The receiver absorbs primary irradiance and the irradiated surfaces transfer heat to the rest of the solid by conduction and re-radiation and to the HTF (air) by convection. The participation of air in radiative transfer in this system is negligible, considering air is mostly transparent in the visible and IR wavelength ranges.

The HFSS used in this study is based on four lamp blocks (light source and reflector) of the German Aerospace Institute (DLR) HFSS as in Fig. 1a. Again, this selection allows for better comparison with literature data. The two blocks placed on the $x-z$ plane are at a 10° angle off the y -axis, while the other two blocks on the $y-z$ plane are at 23.5° off the x -axis. Dimensions and characteristics of the lamp blocks were obtained from reports published by Alkneit, *et al.* [37]. The front of the receiver was placed at the focal point, and the necessary lamp powers were obtained by trial and error till an incident flux of $650[\text{kW}/\text{m}^2]$ was achieved, as reported in the literature work used for verification.

To model the described system, heat, momentum, and radiation transport were considered. However, because a detailed approach to modelling radiative transfer has proven to be critical to model accuracy [38], the geometry is simplified to a single repeating unit or a single channel adiabatic by thermal symmetry (see Fig. 1c). This is a common approach for modelling structured receivers [35] because it allows for a detailed treatment of radiative transfer while lowering the

computational power demand. The next sub-section details the transport equations and boundary conditions comprising the mathematical model.

Details on the transport equations and boundary conditions comprising the mathematical model and results of mesh-independence are available in the [Supplementary Material](#).

With a verified model the next sub-section describes the sensitivity analysis or parametric study used to explore the varied-reflectivity receiver concept.

2.2. Sensitivity analysis & optimization

To study the varied-reflectivity receiver concept, the operating condition, or $P/\dot{m}$ ratio, was fixed at 700 kJ/kg using the same flux characteristics and ambient properties described for the verification case. The receiver's frontal area and channel thickness were, also, fixed at $14 \times 14 \text{ cm}^2$ and 0.4 mm , respectively, to match those of the HiTRec-II.

Fixing $P/\dot{m}$ ratio allows comparing receiver modules under the same operating condition. Meanwhile, fixing channel thickness was mainly done to reduce the number of independent variables without fixing pore size or porosity. Because those two parameters have been noted to have a great impact on receiver's efficiency. Noteworthy, there is a great deal of contradiction in literature on the most favorable combination. Characteristically, Wu, *et al.* [15] and Kribus, *et al.* [39] are examples of

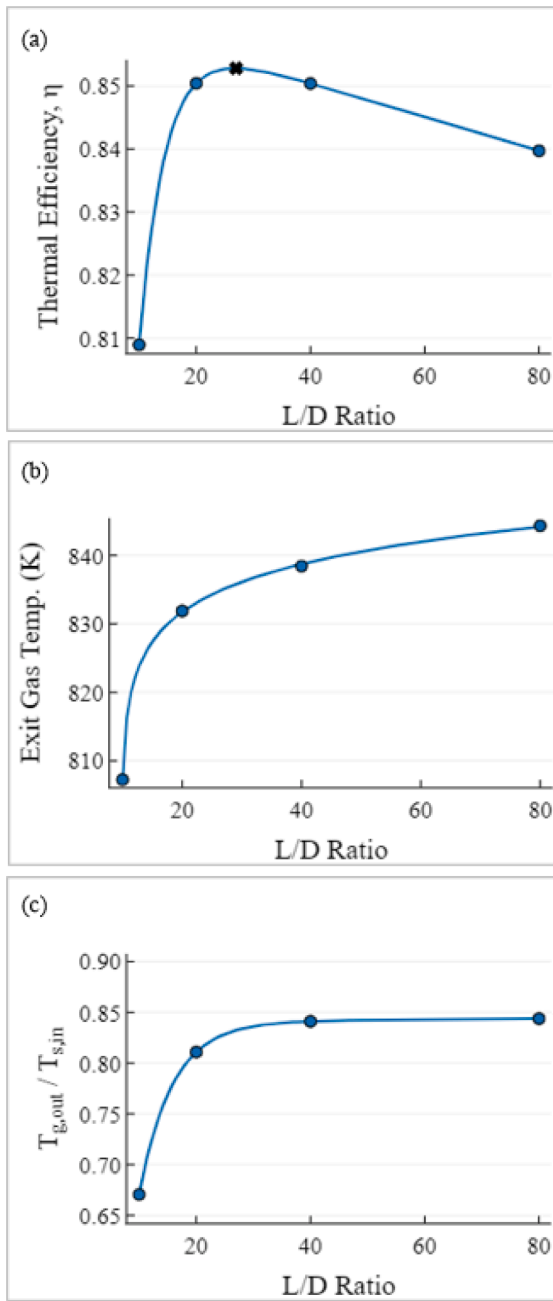


Fig. 6. Sensitivity analysis results for the base case receiver with uniform reflectivity (Table 2) – Stage 2: Effect of length on a) thermal efficiency, b) exit gas temperature, and c) E_{vol} ratio.

Table 3

Optimized receiver designs (for the same average emissivity level, $\varepsilon = 0.8$).

Design Parameter	Base Case	Linear Distribution Case	Wall-Variied Distribution Case
R_{ch} [mm]	1	0.5	0.5
Reflectivity Distribution Variables	$\varepsilon = 0.8$	$L_e/L_{ch} = 0.117$ $m = 1000 [m^{-1}]$	$\alpha_y = 0.9$ $\alpha_x = 0.7$
L_{ch}/D	27.0	18.5	23.4

highly cited works that disagree on the effect of pore size on the thermal efficiency of foam receivers. Additionally, those literature results may not be useful for the varied-reflectivity receivers considered in the current work. One disadvantage of fixing channel thickness is that the

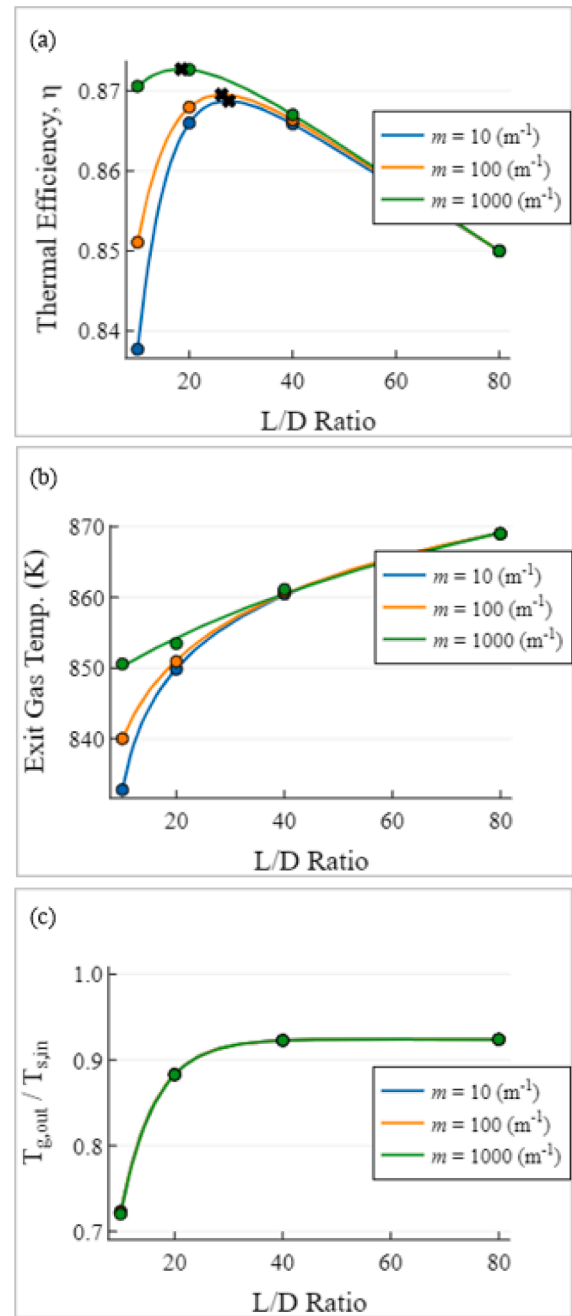


Fig. 7. Sensitivity analysis results for the linear distribution receiver (Table 2) – Stage 2: Effect of length and gradation slope on a) thermal efficiency, b) exit gas temperature, and c) E_{vol} ratio.

effects of porosity and pore size are aliased, but that is addressed later in the discussion section.

After fixing the aforementioned parameters (module area, t_{ch} and P/m ratio), what remains are channel radius (R_{ch}), channel length (or length-to-diameter ratio, L/D) and the reflectivity distribution parameters. Though porosity and pore size, technically, affect both the optical efficiency and HTC of the receiver, their interaction with the reflectivity distribution was expected to be more significant than their interaction with length. Meaning that porosity and pore size were considered key optical parameters in this work and this is discussed further in the following paragraphs.

Meanwhile, channel length is considered typically a heat transfer parameter, as most optically significant phenomena occur in the first few millimeters of the receiver's depth. Any additional length, past the

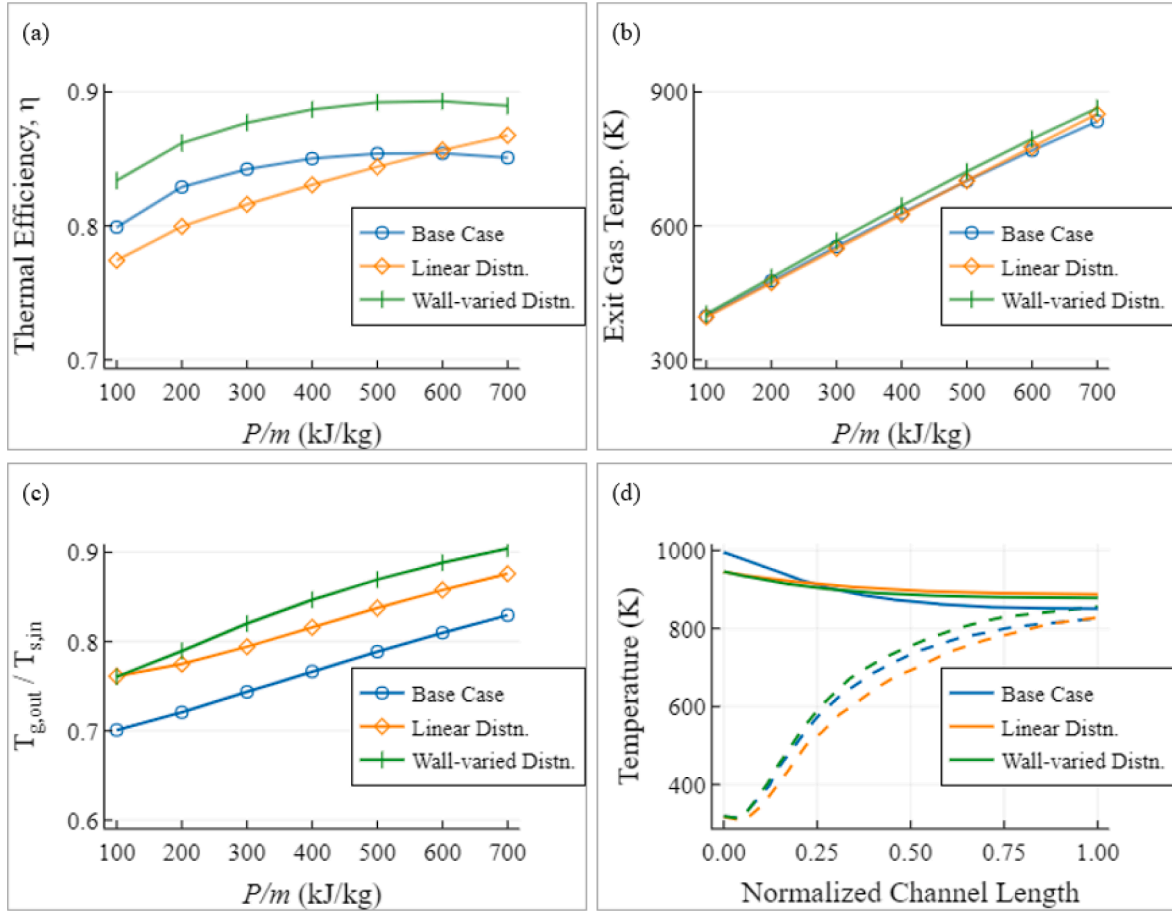


Fig. 8. Performance comparison for the three optimized receiver designs in terms of a) thermal efficiency, b) exit gas temperature, c) the E_{vol} ratio, and d) axial temperature profiles against normalized length for the three receivers at $P/\dot{m} = 700$ [kJ/kg].

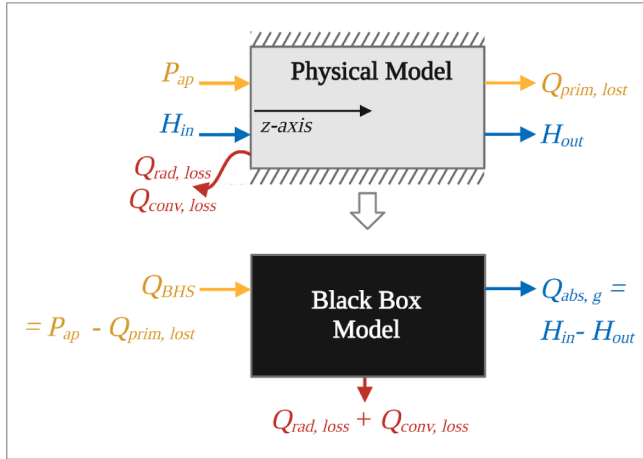


Fig. 9. Overall energy balance on a volumetric receiver module and resulting energy balance components used for analysis.

extinction and absorption of intercepted light rays, has a greater impact on the allotted gas–solid contact time and overall heat transfer coefficient of the receiver. This assumption for the effect of length could be verified by checking the calculated axial heat source distribution and temperature profiles. Length does not play an important role in radiation absorption, if the heat source reaches zero before the exit of the receiver, and the gas and solid temperatures reaches thermal equilibrium ($T_{g,out} \approx T_{s,out}$) by the exit of the receiver. On the other hand, the channel

length could become a significant optical parameter if radiation penetration is significantly improved, which leads to main objective of this work. Hence, the above verification tests were conducted for the reflective receiver cases.

According to the above assumptions, the general optimization procedure for the receiver variations considered in this work consisted of two stages: i) obtain the optimal combination of key optical properties (R_{ch} or ϕ and $\varepsilon(x, y, z)$), and ii) optimize key thermal parameters (or channel length to diameter ratio). The optimization criterion was to maximize thermal efficiency for any given parametric sweep or stage, where thermal efficiency is given by:

$$\eta = \frac{\dot{m}(C_{p|g,out}T_{g,out} - C_{p|g,in}T_{g,in})}{P_{ap}} \quad (1)$$

Exit gas temperature, $T_{g,out}$, and the volumetric effect were also monitored as relevant performance parameters. However, since there are no formally agreed upon quantitative definitions for the volumetric effect, E_{vol} , used by [30] was employed:

$$E_{vol} = \frac{T_{g,out}}{T_{s,in}} \quad (2)$$

Wherein, a level of the volumetric effect is achieved when $E_{vol} > 1$. The volumetric effect and possible definitions are further explored in dedicated discussion section.

The reflectivity distributions considered in this work were an axial linear distribution as in Fig. 2a and a wall-varied reflectivity distribution as in Fig. 2b. The linear distribution has a highly reflective region in a front region, entry length L_e , to reflect primary rays deeper into the receiver leading to a more uniform heat source distribution. The wall-

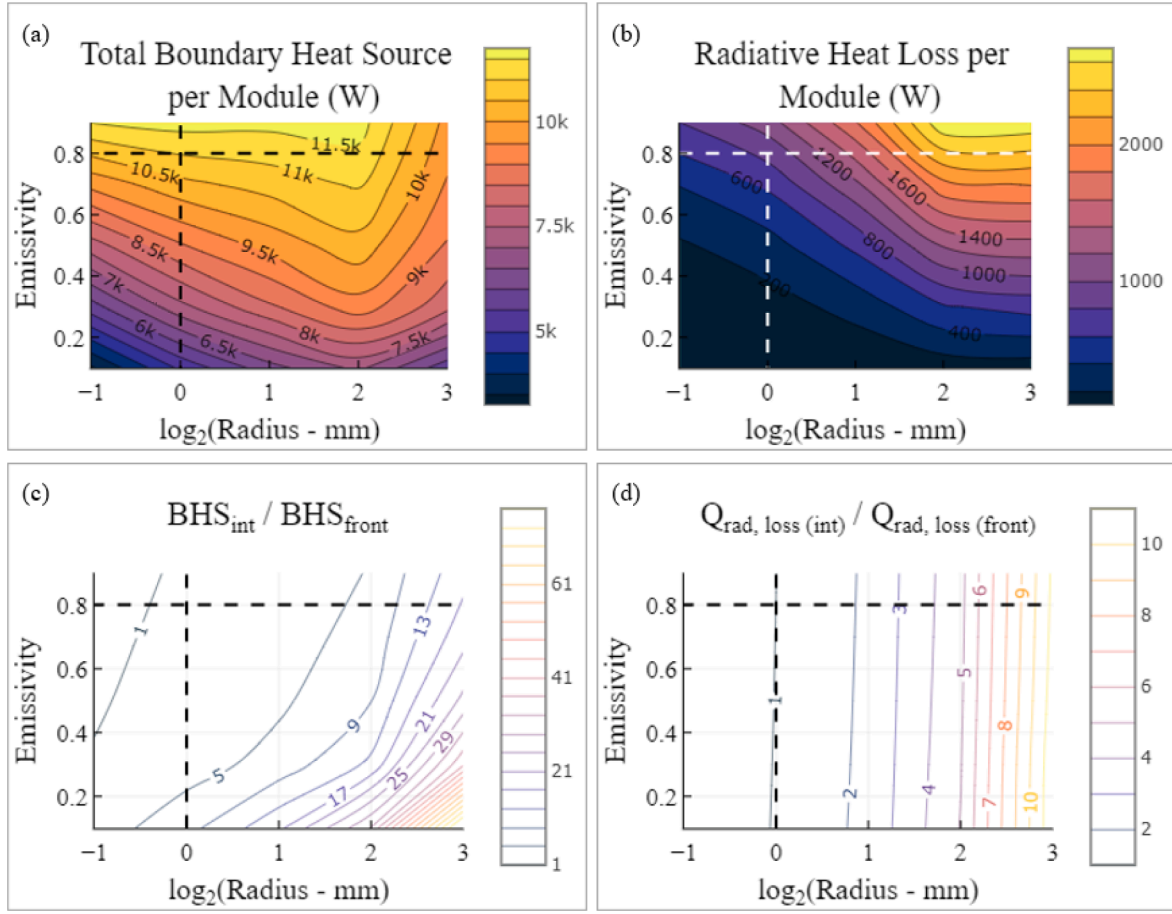


Fig. 10. Energy balance components for base case receiver in terms of a) total BHS per module, b) radiative heat loss per module, c) ratio of BHS on interior to front surfaces, and d) ratio of $Q_{\text{rad, loss}}$ from interior to front surface (dashed black/white lines indicate the receiver design corresponding to peak efficiency).

varied distribution has a higher reflectivity on two of its interior walls to increase primary ray reflections and a low reflectivity on the other two to absorb the secondary and incident rays. This latter design was considered because it should be compatible with more manufacturing techniques compared to the linear distribution design.

For both of these distributions, the maximum absorptivity limit, α_{high} , was applied to the front wall of the channel (as in Fig. 2b), since increasing the reflectivity on that surface would not improve radiation penetration but only increase optical losses. To put the results of these two designs into context, an additional receiver with uniform reflectivity was considered as the base case.

Because both the varied-reflectivity receiver cases have more than one distribution parameter, some additional considerations were made to simplify optimization. For the linear distribution case, in the first stage for optical parameter optimization, only entry length and channel radius were varied using a very high gradation slope ($m = 1000\text{m}^{-1}$) creating an almost stepped distribution (see green line in Fig. 2a). For the wall-varied distribution case, each set of parallel walls receives radiation at a different off-axis angle. Hence, the effect of off-axis angle was studied (results not shown), and it was observed that applying the high absorptivity limit to the walls receiving radiation at a steeper angle was significantly more advantageous. Therefore, the absorptivity of the y-walls, α_y (Fig. 2b), was fixed to the limit α_{high} . Table 2 shows the levels used for each of the parametric sweeps.

3. Results

This section summarizes the results of the sensitivity analysis and optimization studies for the two optimization stages: optical parameter

optimization, and thermal parameters or length optimization. Interested readers can check the verification results obtained for selected literature cases in the [Supplementary Material](#).

3.1. Effect of key optical properties (Average emissivity and Porosity)

Fig. 3 shows the results for the first stage of optical property optimization and an uniform reflectivity across the receiver (base case - Table 2). The thermal efficiency (Fig. 3a) increases monotonically with emissivity for any given porosity-radius pair, but for a given emissivity there is a porosity-radius pair that maximizes thermal efficiency. Exit gas temperature (Fig. 3b) exhibits the same trend as thermal efficiency, hence, its variation with emissivity and porosity-radius pairs have been omitted for the remaining receiver cases for the sake of brevity. Literature thrives with works aiming for the volumetric effect because of the aforementioned advantages; in contrast to intuition Fig. 3c the maximum E_{vol} ratio corresponded to a region of very poor performance. Similar observations were made by others (e.g., [30]) when using the same volumetric effect definition. This implies that the quantitative definition (eqn. (2)) cannot provide a holistic representation of the volumetric effect. Eventually, the latter should not had been a surprise since this definition only accounts for the inlet and outlet temperatures, ignoring for example the formation of hotspots near the inlet or the uniformity of heat source distribution. Despite the inconsistency and the later exploration in Discussion, the ratio $E_{\text{vol}} = T_{\text{g, out}} / T_{\text{s, in}}$ (eqn. (2)) is reported for subsequent receiver cases for the sake of comparison.

Continuing the analysis of the optical properties' optimization (Fig. 3) for the base case receiver, as expected, the highest thermal efficiency is achieved at the highest emissivity. Hereafter, the performance

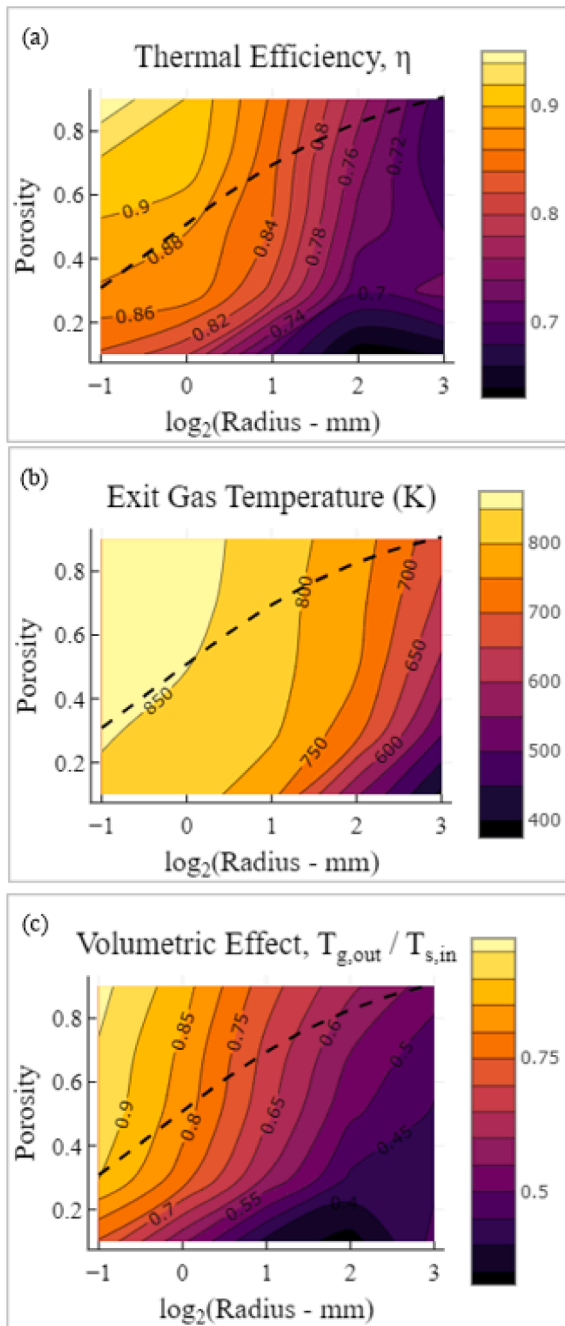


Fig. 11. Parametric study results for effect of channel radius and porosity on the base case receiver's a) thermal efficiency, b) exit gas temperature, and c) E_{vol} ratio (dashed black lines indicate designs where $t_{ch} = 0.4$ mm, i.e. considered designs in all other receiver cases presented in this work).

of the remaining receiver designs is compared at the selected emissivity levels of SiSiC, around 0.8. At that emissivity, the porosity-radius pair that maximizes thermal efficiency is 0.51 and 1 mm, respectively.

For the linear distribution case (Table 2) a different behaviour is noted (Fig. 4) compared to the base case. Thermal efficiency (Fig. 4a) and exit gas temperature (not shown herein) no longer exhibit a maximum with porosity and radius, and for any given emissivity, the smallest porosity and radius will maximize thermal efficiency. Additionally, the ratio E_{vol} becomes less dependent on average emissivity. At the average emissivity of 0.8, the porosity-radius pair corresponding to maximum thermal efficiency is 0.31 and 0.5 mm.

A similar behaviour is observed for the wall-varied distribution case

(Table 2). Wherein, thermal efficiency (Fig. 5a) and exit gas temperature (not shown herein) do not exhibit a maximum with porosity and radius and the volumetric effect ratio becomes even less dependent on average emissivity (Fig. 5b). At the average emissivity of 0.8, the porosity-radius pair corresponding to maximum thermal efficiency is, also, 0.31 and 0.5 mm.

Important to note that the results shown in Fig. 5 are for the optimized absorptivity ($\alpha_y = 0.9$) for the off-axis angles of incident rays; not shown here for the sake of brevity.

3.2. Effect of key thermal parameters (Length to diameter Ratio)

Fig. 6 presents the receiver's length optimization results for the base-case (Table 2) upon fixing the optical parameters to their previous obtain optimum values. The thermal efficiency exhibits a maximum with length, while exit gas temperature and the ratio E_{vol} increase monotonically with length till they reach a plateau. The length to diameter ratio that maximizes thermal efficiency is 27, producing a receiver with the dimensions: $R_{ch} = 1$ mm, $t_{ch} = 0.4$ mm, and $L_{ch} = 54$ mm. These are very similar to the dimensions of the current state-of-the-art HiTRec-II receiver (with $R_{ch} = 1$ mm, $t_{ch} = 0.4$ mm, and $L_{ch} = 50$ mm), which further verifies the developed model. The slightly longer optimum length of the receiver, in this work, was possibly due to the steeper off-axis angles of the light sources used in this study.

The same trends are observed for the varied reflectivity cases (not shown). Wherein, thermal efficiency goes through a maximum with increased length, while exit gas temperature and the ratio E_{vol} increase monotonically till they reach a plateau. The length to diameter ratios that maximize thermal efficiency for these two designs are summarized in Table 3.

Additionally, Fig. 7 shows that the effect of the gradation slope (linear distribution case in Table 2) on the volumetric effect (Fig. 7c) is negligible, and on performance (Fig. 7a) and exit gas temperature (Fig. 7b) is minimal and somewhat negative. Hence, there is no advantage to a gradual changing reflectivity compared to a step change (gradation slope $m = 1000\text{m}^{-1}$).

3.3. Comparing optimized receivers

The optimized design parameters for each of the studied receiver cases are summarized in Table 3 and the performance charts and temperature profiles are plotted in Fig. 8 for each mass rate specific powers. The latter were produced by varying the inlet mass flowrate into the receiver for the same power on aperture to explore how the receivers' performance are affected under a range of $P/\dot{m}$ ratios, similar to the ones reported by Capuano, *et al.* [31] for varied porosity receiver.

The thermal efficiency (Fig. 8a) goes through a maximum around $P/\dot{m} = 600$ kJ/kg for the base case and wall-varied distribution cases. The efficiency of the linear distribution case appears to reach a maximum outside of the considered operating range. Meanwhile, exit gas temperature (Fig. 8b) and the ratio E_{vol} (Fig. 8c) increase monotonically with $P/\dot{m}$ ratio, due to the increased residence time (increasing $P/\dot{m} \rightarrow$ decreasing mass flowrate $\rightarrow$ lower gas velocity). Additionally, the wall-varied distribution case exhibits improved performance compared to the base case with a maximum of about 0.89, while the linear distribution case only exceeds the base case efficiency at higher $P/\dot{m}$ ratios, reaching 0.87 at $P/\dot{m} = 700$ kJ/kg. Equally importantly, both receivers with varied-reflectivity achieve a flatter solid temperature profile and improved volumetric behavior compared to the base case at $P/\dot{m} = 700$ [kJ/kg].

The reasons for improved performance and for the changed shape of the temperature profiles are discussed in the following section.

4. Discussion

This chapter starts by exploring and inferring the reasons behind the

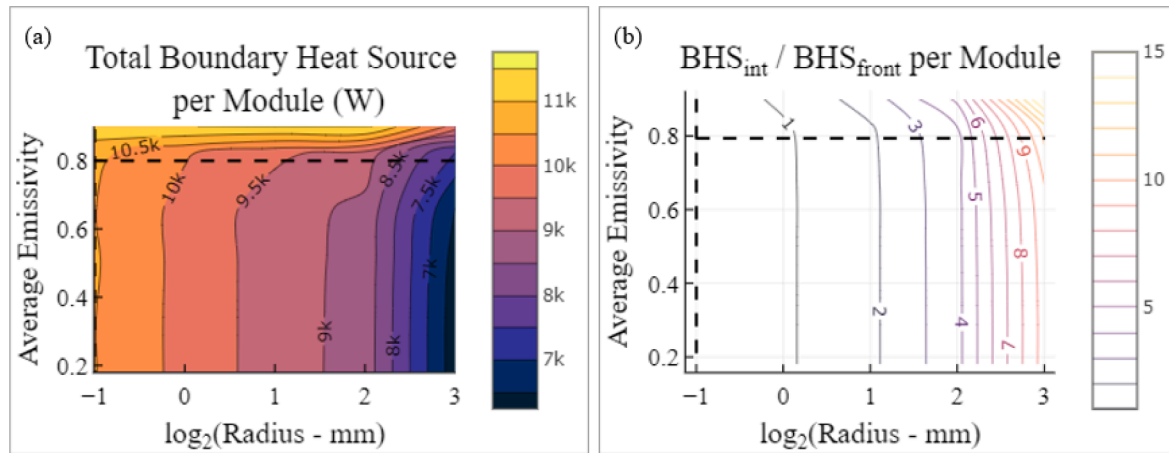


Fig. 12. Energy balance components for linear distribution case receiver in terms of a) total BHS per module, and b) ratio of BHS on interior to front surfaces (dashed black/white lines indicate the receiver design corresponding to peak efficiency).

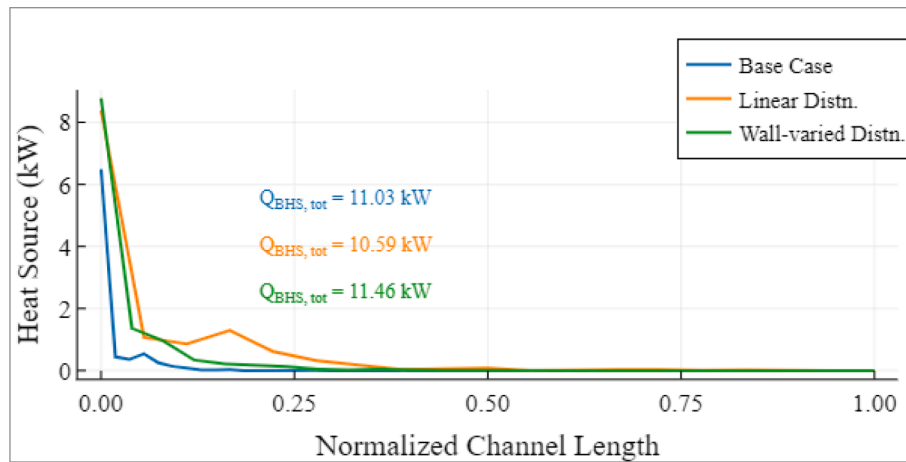


Fig. 13. Axial boundary heat source distribution for each optimized receiver module with total heat sources annotated on the figure.

slightly improved performance of the varied reflectivity receivers presented so far. In particular, the focus is on the energy balance components of each receiver design and how those affect the receiver's performance, and some parameter aliasing is addressed. As an outcome, an additional varied-reflectivity receiver that utilizes specular reflectivity was devised and the chapter continues demonstrating the specular receiver's significantly improved volumetric behavior and performance. Finally, the chapter completes with an exploration and comparison of potential quantitative definitions of the volumetric effect, motivated simultaneously by its importance and weak literature definition of the volumetric behavior.

4.1. Analysis of energy balance components

The trends observed in the sensitivity analysis section can be explained using the overall energy balance components which are depicted in Fig. 9 for the open-air volumetric receiver.

Plotting the contours for these energy balance components in Fig. 10, one can infer that the total boundary heat source (BHS; see Fig. 10a) increases monotonically with emissivity and goes through a maximum at a pore radius around $R_{ch} = 1 \text{ mm}$ – and porosity at around $\phi = 0.69$. At lower porosities and channel radii, poor radiation penetration limits the boundary heat source, while at higher porosities and channel radii, the decrease in surface area and pore velocity limit radiation absorption and HTC. Additionally, the effect of decreased porosity and channel radius on total BHS becomes weaker at higher emissivity, implying that

the majority of radiation ends up being absorbed by the front surface of the receiver (see Fig. 10c). Furthermore, radiative losses to the ambient are larger for larger channel sizes and emissivity (see Fig. 10b), due to the increased ambient view factors and the surface area participating in radiative transfer. This pushes the maximum efficiency towards lower porosity-pore radius pairs at higher emissivity.

Favouring lower porosities is not a common conclusion in literature, as most works agree that higher porosity is favourable because it increases radiation penetration and, in turn, the total BHS. The effects of porosity observed in this work are most likely a by-product of the study design. Channel thickness was fixed in all the parametric studies conducted, which aliases the effects of pore size and porosity, as mentioned previously in section 2.2. The individual effects of porosity and pore size can be observed by conducting a parametric sweep for the base case receiver where only R_{ch} and ϕ are varied, while the length-to-diameter ratio, surface emissivity and $P/\dot{m}$ ratio are fixed at 25, 0.8, and 700 kJ/kg, respectively. The results of this sweep (Fig. 11) show that higher porosities and smaller channel sizes are most favourable, but none achieves the volumetric effect in terms of the ratio E_{vol} , contrary to conclusions by Wu, *et al.* [15].

Nevertheless, comparing the energy balance components for the base case in Fig. 10a to those of the varied-reflectivity receiver cases in Fig. 12a, total BHS no longer exhibits a definite maximum with porosity and channel radius, but instead it exhibits a bimodal behaviour. At high emissivity levels ($> \sim 0.75$), the effect of porosity and channel radius on BHS becomes less pronounced. In contrast, minimizing porosity and

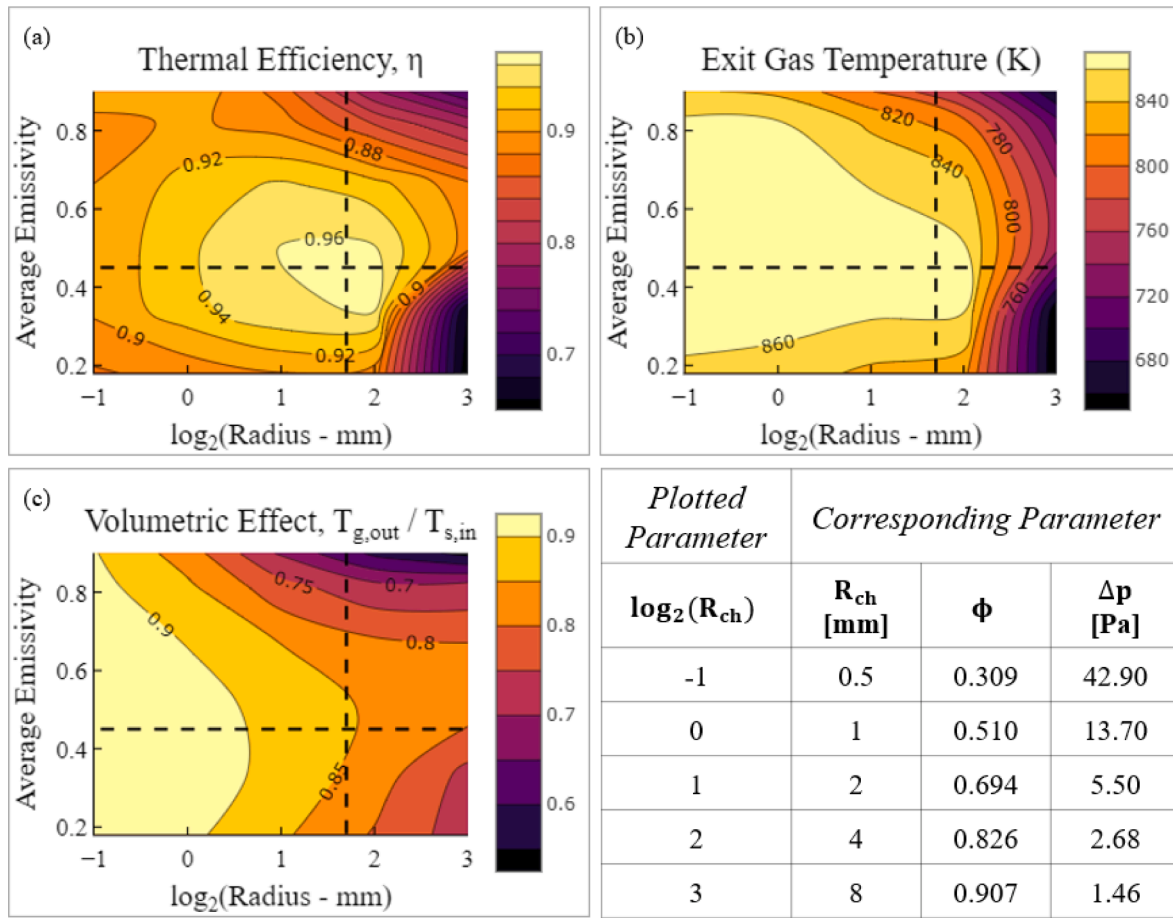


Fig. 14. Specular case receiver optimization – Stage 1: Effect of emissivity and porosity (dashed black lines indicate the selected receiver design). Shown are contours for a) thermal efficiency, b) exit gas temperature, c) E_{vol} ratio.

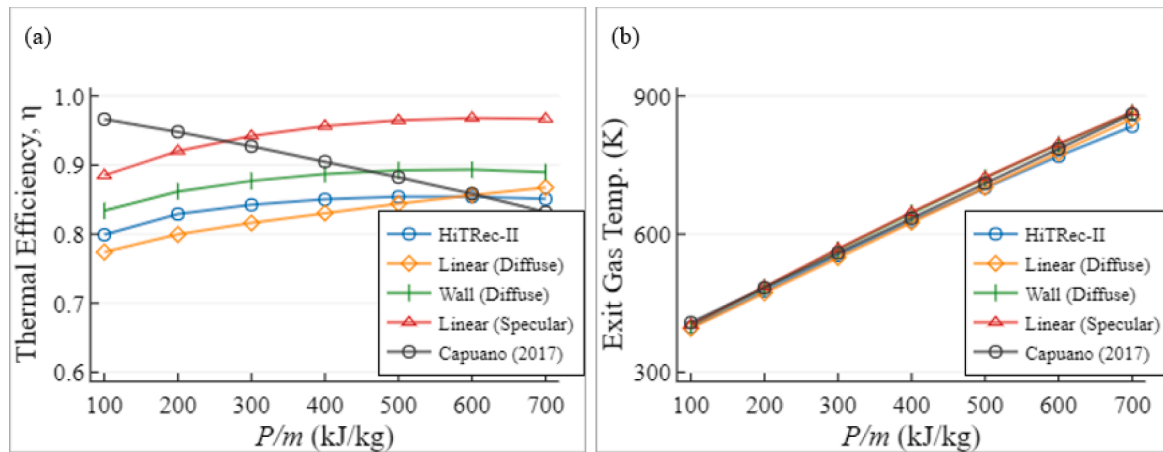


Fig. 15. Performance comparisons for HiTRec-II, optimized diffuse and specular varied reflectivity receivers, and the varied porosity receiver designed by Capuano, et al. [31] in terms of a) thermal efficiency and b) exit gas temperature.

channel radius becomes advantageous for all emissivity values less than ~ 0.75 . In other words, the effect of average emissivity on BHS becomes less noticeable for lower average emissivity or longer entry region length. These results are counter-intuitive, as lowering emissivity should have a greater effect on the boundary heat source. This implies that the improvement in boundary heat source comes mostly from increased absorption by the front surface of the receiver (or improved optical efficiency) rather than improved radiation penetration. Fig. 12b shows

that the optimized receiver is in a region where the majority of primary ray absorption is done by the front surface due to the high emissivity applied there.

This is further proven by plotting in Fig. 13 the boundary heat source distribution for each optimized receiver case examined herein. There is no clear improvement in the boundary heat source distribution between the test cases, only an increase in its total value. The same figure, also, illustrates the inadequacy of the employed definition for the volumetric

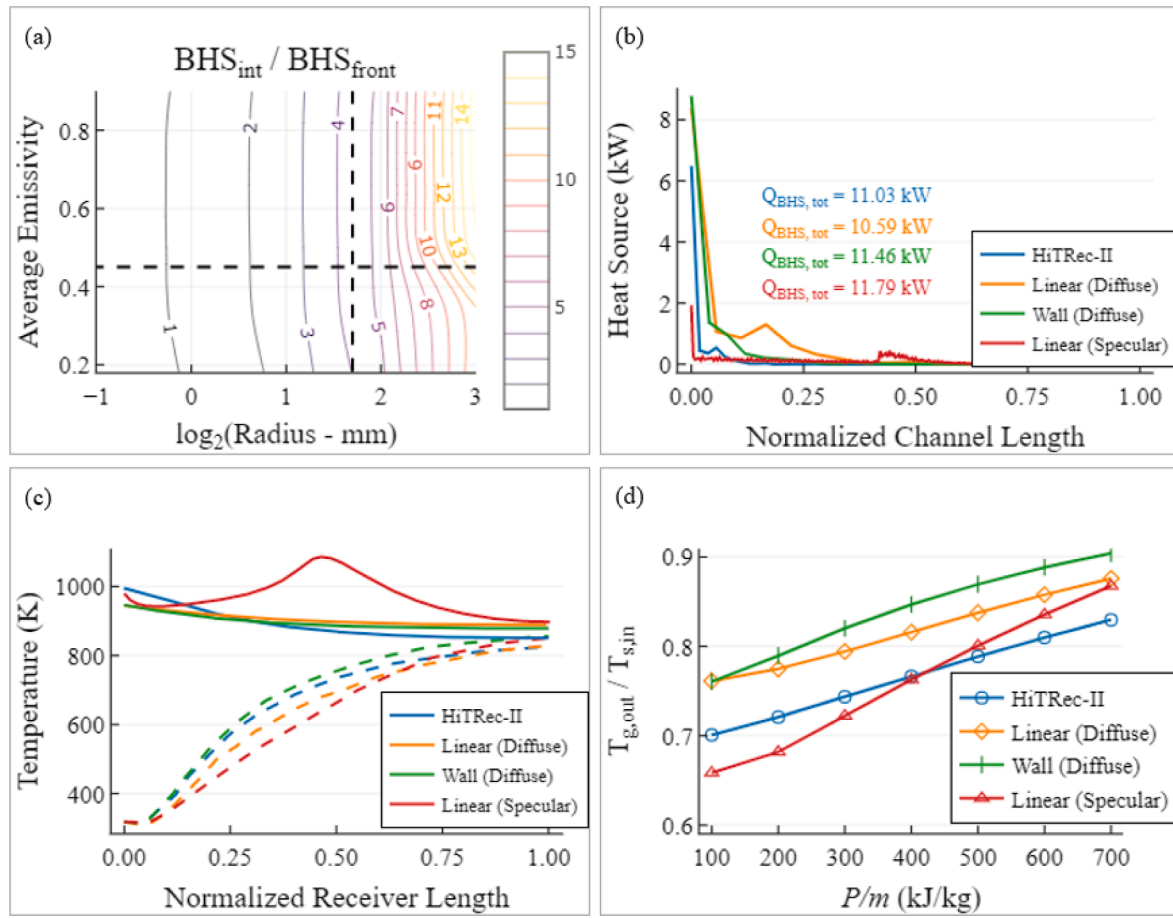


Fig. 16. Examples of specular varied-reflectivity receiver exhibiting volumetric effect in terms of: a) ratio of BHS on interior to front surfaces (dashed black lines indicate the receiver design corresponding to peak efficiency), b) axial BHS profiles, c) axial temperature profiles at $P/\dot{m} = 700 \text{ kJ/kg}$, and d) E_{vol} ratio.

effect to capture the aforementioned phenomena. Wherein, the varied-reflectivity receivers exhibited an improved volumetric effect ratio, but there is little to no difference between the designs in radiative penetration, or full volumetric utilization. This highlights the need for an exploration of more descriptive and accurate quantitative definitions of the volumetric effect, which is presented in section 4.3.

Additionally, Fig. 13 implies that adding a reflective region is only slightly advantageous to radiative penetration, which is contrary to the authors' intuition and to previous qualitative findings by [34]. This led to a second attempt using a specularly reflective surfaces instead of diffuse ones, which is discussed in section 4.2 below.

4.2. Specular Varied-Reflectivity receiver case study

To explore the effect of a specular reflective region, the same sensitivity analysis conditions mentioned in Table 2 for stage 1 of the linear distribution case were applied, with the reflective region being fully specular and the high absorptivity region being fully diffuse. The results for stage 1 of this analysis are shown in Fig. 14.

The results show a completely different behaviour to the (fully diffuse) linear distribution case examined earlier, with the thermal efficiency and exit gas temperature no longer exhibiting the same trends. In particular, thermal efficiency exhibits a maximum around $(R_{ch}, \bar{\epsilon}) = (3.2 \text{ mm}, 0.45)$ and drops dramatically at higher porosities and channel sizes. Peak efficiency corresponds to a maximum for exit gas temperature, except that at lower porosities and channel sizes, the design parameters (R_{ch} and $\bar{\epsilon}$) have minimal impact on the exit gas temperature. The ratio E_{vol} exhibits a maximum around $\bar{\epsilon} = 0.45$, but monotonically decreases with porosity and channel size.

This design achieves better performance compared to all other fully diffuse receiver cases, and comparable performance to the graded porosity receiver designed by Capuano, *et al.* [31] (Fig. 15). In fact, the specular receiver maintains a high performance for all considered $P/\dot{m}$ ratios, while the efficiency of the varied porosity receiver by Capuano, *et al.* [31] drops significantly at higher $P/\dot{m}$ ratios.

Also, the specular receiver exhibits improved volumetric effect illustrated by the boundary heat source distributions shown in Fig. 16a & b and the temperature profiles in Fig. 16c. The axial BHS profile, also, confirms the assumption made during study design, which is that channel length is a more significantly heat transfer parameter than an optical parameter even when radiation penetration is improved. For the specular case shown in the figure ($R_{ch} = 4 \text{ mm}$, $L_{ch} = 200 \text{ mm}$, $L_e = 100 \text{ mm}$), increasing the channel length would have allowed the solid and gas temperatures to reach equilibrium, but it would not have increased the amount of ray power absorbed significantly.

Specular reflectivity is not very common among typical CSP receiver materials, but there are commercially available processes for producing mirrors out of common OVAR materials like SiC and SiSiC [40]. Other candidate materials are MAX-phase ceramics. These are a class of materials that inherit the high thermal and chemical stability of ceramics, combined with the high conductivity and (more importantly) polishability of metals. Gonzalez-Julian [41] suggested that Ti_2AlC and Cr_2AlC had promising thermal and optical performances to be used in CSP. Indeed, the stable optical properties for Cr_2AlC after exposure to concentrated radiative flux reported by Sarwar, *et al.* [42] were $\alpha = 0.15$, which is very close to the $\alpha_{low} = 0.10$ used to produce the specular receiver results above. Hence, there is apt potential for the manufacturability of the specular receiver in terms of material, but some

Table 4
Considered quantitative definitions for the volumetric effect.

Definition	Equation	Target
Basis: A volumetric receiver should exhibit a frontal solid temperature lower than the exit gas temperature Volumetric Effect Ratio*	$E_{vol} = \frac{T_{g,out}}{T_{s,in}}$	$E_{vol} > 1$
Basis: If a volumetric receiver is compared to a heat exchanger, the same performance criteria can be used LMTD	$\frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1 / \Delta T_2)}$	$\max(LMTD)$
Basis: Achieving the volumetric effect should minimize radiative losses to the ambient and improve heat source distribution Ratio of radiative loss to the heat absorbed by the gas	$\frac{Q_{rad,loss}}{Q_{abs,g}}$	$\min\left(\frac{Q_{rad,loss}}{Q_{abs,g}}\right)$
Fraction of radiative loss from the total heat source	$\frac{Q_{rad,loss}}{Q_{BHS}}$	$\min\left(\frac{Q_{rad,loss}}{Q_{BHS}}\right)$
Basis: An improved volumetric effect would ensure a favorable boundary heat source distribution and in turn maximize the portion of the boundary heat source absorbed by the gas Ratio of heat absorbed by the gas to the total heat source	$\frac{Q_{abs,g}}{Q_{BHS}}$	$\frac{Q_{abs,g}}{Q_{BHS}} = 1.0$
Basis: Achieving the volumetric effect should improve the optical efficiency Fraction of total heat source from power on aperture (Optical Efficiency)	$\frac{Q_{BHS}}{P_{ap}}$	$\frac{Q_{BHS}}{P_{ap}} = 1.0$
Basis: An ideal receiver should have a local temperature driving force of zero. This can be evaluated using statistical definitions RMSE	$\sqrt{\frac{1}{N} \sum_i (T_{s,i} - T_{g,i})^2}$	$RMSE = 0$
NMB	$\frac{\overline{T_s} - \overline{T_g}}{\overline{T_g}}$	$NMB = 0$
NMSD	$\frac{\sigma_g - \sigma_s}{\sigma_g}$	$NMSD = 0$
Basis: Full utilization of the entire receiver volume requires a uniformly distributed driving force Standard deviation in ΔT driving force ($\sigma_{\Delta T}$)	$\sqrt{\frac{1}{N} \sum_{i=1}^N (\Delta T - \overline{\Delta T})^2}$	$\min(\sigma_{\Delta T})$

* Previously used by Luque, *et al.* [30].

experimental work on validating the above claims and correctly assembling the receiver are necessary.

Furthermore, Fig. 16d shows the volumetric effect in terms of the ratio $E_{vol} = T_{g,out}/T_{s,in}$. This definition of the volumetric effect implies that the specular receiver has poorer volumetric behaviour compared to the two diffuse varied-reflectivity receivers, despite the specular receiver having a much more uniform BHS distribution. This further highlights the need for more holistic quantitative definitions for the volumetric effect, which is addressed in the following section.

4.3. An exploration of quantitative definitions of the volumetric effect

Quantitative definitions for the volumetric effect explored in this study were collected from past literature or devised based on intuition and relevant statistical metrics and then grouped by the basis of their use (Table 4). One way to approximate the efficacy of these definitions is by whether they follow the trend of thermal efficiency. This was done by comparing the different volumetric effect definitions against thermal efficiency for a single receiver under different $P/\dot{m}$ ratios, in which case the volumetric effect would be seen as an operational parameter. Alternatively, it can be seen as a design criterion assisting on the selection or optimization of the candidate designs. This was done by first using the different definitions to optimize the selected receivers, and then comparing the thermal performance of the optimized receivers, all operated under the same $P/\dot{m}$ ratio. The next two sections explain in detail the two ways approximating the efficacy of the volumetric definitions.

4.3.1. Screening volumetric effect definitions at different operational conditions

Fig. 17 shows the definitions introduced in Table 4 for the optimized

base case receiver operated at various $P/\dot{m}$ ratios (100 to 700 kJ/kg). Definitions unbounded by upper and lower limits (i.e., 0 to 1 or -1 to 1) were normalized within their respective datasets according to:

$$\hat{x}_i = \frac{x_i}{\max(|X|)} \quad (3)$$

The LMTD, the earlier employed ratio $E_{vol} = T_{g,out}/T_{s,in}$, and the driving force non-uniformity ($\sigma_{\Delta T}$) have similar trends (Fig. 17a and d) and approach the same global maximum thermal efficiency but at a volumetric effect value slightly lower than their theoretical targets. Defining the volumetric effect based on driving force non-uniformity (σ_{DF} shown in Fig. 17d) achieved a similar undesirable trend.

From the definitions based on power ratios (Fig. 17b), only $Q_{abs,g}/Q_{BHS}$ points to the maximum thermal efficiency, which is expected as it shares the same term ($Q_{abs,g}$) with the thermal efficiency's equation and Q_{BHS} is a constant for all the cases presented in this section. Hence, this definition is somewhat trivial. The Q_{BHS}/P_{ap} is, also, trivial for the same reason without even converging to the proper thermal efficiency. The remaining definitions ($Q_{rad,loss}/Q_{BHS}$ and $Q_{rad,loss}/Q_{abs,g}$) have the appropriate trend pointing to a global optimum thermal efficiency but not at their own target values. This is caused by the high solid temperatures and radiative losses at high power on aperture to mass flowrate ratios.

Finally, of the definitions based on statistical quantities (Fig. 17c), only the normalized mean standard deviation (NMSD) reaches its target value of zero, but that does not correspond to the maximum efficiency. Meanwhile, RMSE and NMB do not reach their targets, with the thermal efficiency decreasing significantly while they approach zero.

The reason many of these definitions accomplish the same trends but fail to capture the optimum thermal efficiency is likely because the temperature profiles achieved are more or less the same for all tested $P/\dot{m}$ ratios (Fig. 18). This implies that defining the volumetric effect as an operational parameter is not a good way for screening definitions. Instead, the next section applies the same definitions considered in Table 4 to different receiver designs operating under the same $P/\dot{m}$ ratio.

4.3.2. Screening volumetric effect definitions for different designs

In this case, different volumetric effect definitions are plotted against key optical properties like the contour plot for $T_{g,out}/T_{s,in}$ in Fig. 3c). This produces an issue of what designs to select to compare different volumetric effect definitions. Plotting η vs. the volumetric effect for a cutline or set of receiver designs like the one that maximizes thermal efficiency (see Fig. 19) proved to be misleading, as certain definitions will follow the trend of thermal efficiency along the cutline, but not elsewhere in the design space. One example is the volumetric effect ratio ($T_{g,out}/T_{s,in}$) along the cutline shown in Fig. 19 for the base case. Along the cutline – as in Fig. 20 – $T_{g,out}/T_{s,in}$ follows the trend of thermal efficiency. However, over the entire $R_{ch-\epsilon}$ design space, the design that enhances $T_{g,out}/T_{s,in}$ corresponds to poor thermal efficiency, which was the original motivation behind this investigation.

To eliminate the bias added by selecting a set of designs to compare definitions, a search for the extrema of volumetric effect definitions instead of trends was carried out. The result of this search for the base case is Fig. 21.

For the base case receiver (Fig. 21a), most of the volumetric effect definitions leaned towards the design with the lowest emissivity and lowest radius (porosity). The former observation agrees with previous works [39] but the latter (low porosity leading to enhanced volumetric effect) contradicts previous literature and intuition. This is likely a byproduct of the study design used here, where the lowest porosity designs corresponded to the smallest channel radii, and, in turn, the highest interstitial velocities. The definition favoring a design closest to the one maximizing efficiency is Q_{abs}/Q_{BHS} . This is not a trivial result, as the different cases considered here have different total heat sources (Q_{BHS}).

The same analysis was conducted on the specular receiver case which

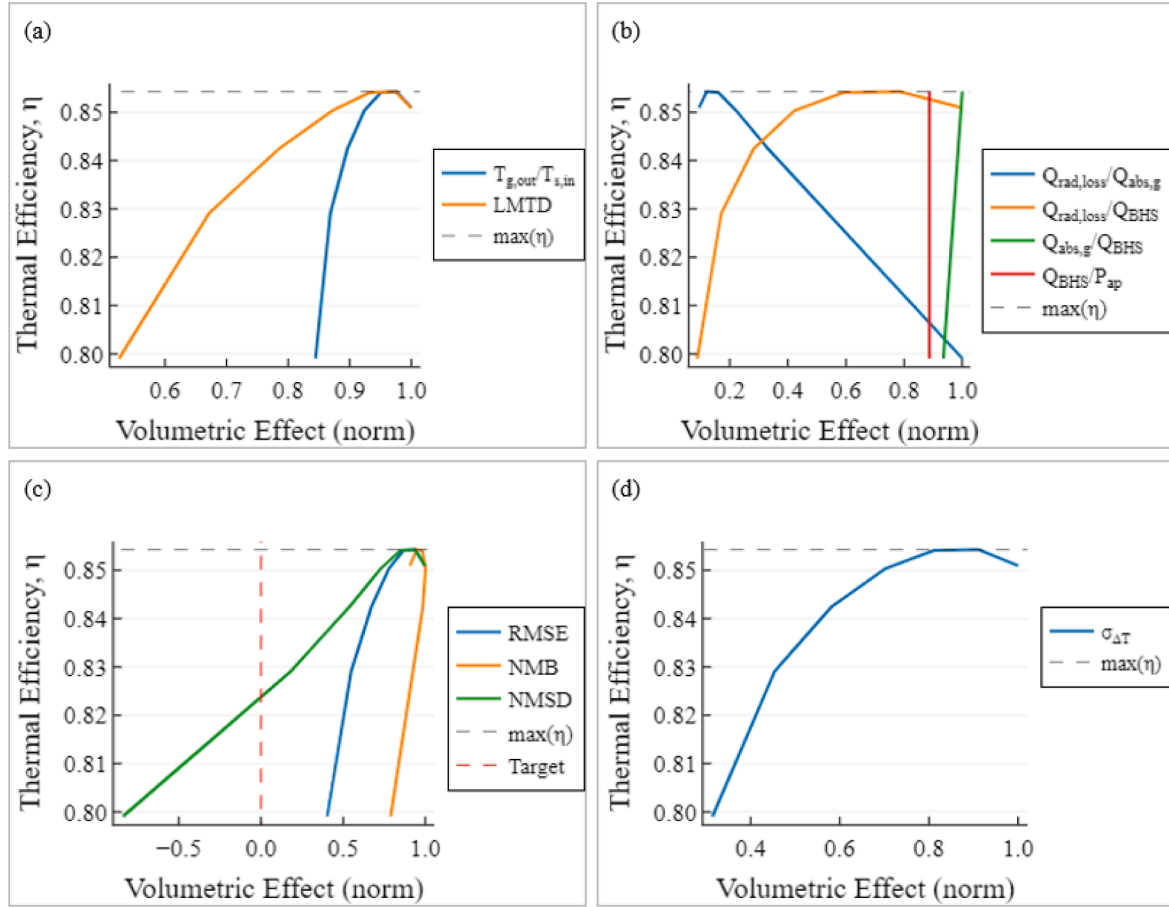


Fig. 17. Volumetric effect definitions as operational parameters based on: a) inlet and outlet temperatures, b) power ratios, c) statistical evaluations of the local driving force, and d) driving force non-uniformity; applied to the optimized base case receiver ($R_{ch} = 1\text{ mm}$, $L/D = 25$, $t = 0.4\text{ mm}$) at $P/\dot{m}$ ratios ranging from 100 to 700 kJ/kg.

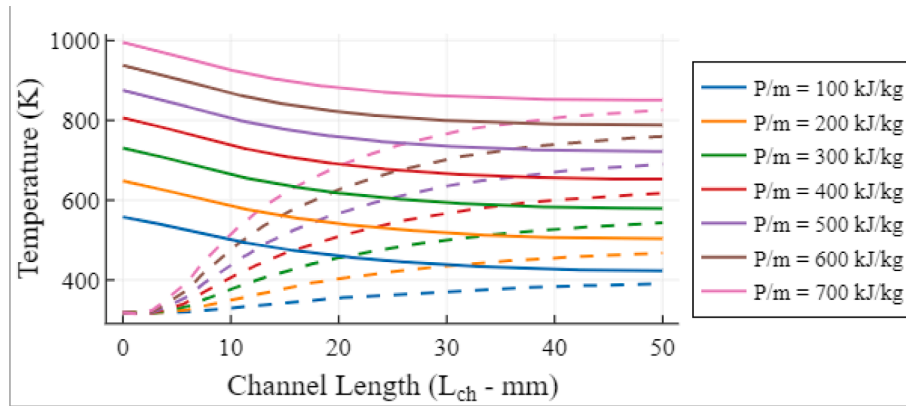


Fig. 18. Axial temperature profiles for the optimized base case receiver ($R_{ch} = 1\text{ mm}$, $L/D = 25$, $t = 0.4\text{ mm}$) operated at $P/\dot{m}$ ratios ranging from 100 to 700 kJ/kg.

clearly exhibited volumetric behaviour (described in section 4.2). The targeted designs for each definition are identified in Fig. 22a along with the achieved thermal efficiencies Fig. 22b. Interestingly, each definition targeted a unique receiver design, though most of them achieved similar efficiencies.

Both definitions that are based on the inlet and outlet temperatures ($T_{g,out}/T_{s,in}$ and $LMTD$) favored designs with low average emissivity achieving $\eta_{th} = 0.9$. However, these designs are not equivalent. Examination of the axial temperature profiles for these designs (Fig. 23a), showed that $LMTD$ favours large channel sizes and a lower emissivity

because those produce the largest driving forces at the back of the receiver. Meaning, if the design (or $R_{ch} - \bar{\epsilon}$ pair) that maximized $LMTD$ was taken to the next stage of length optimization where the gas and solid temperatures were allowed to reach equilibrium at the back, $LMTD$ would no longer favour that $R_{ch} - \bar{\epsilon}$ pair. This effect is more severe in the base case where the temperature profiles of almost all designs reach equilibrium at the back (figure not shown). There, $LMTD$ is maximized for designs with the largest temperature driving force in the front or the highest frontal surface temperature. Meanwhile, the ratio $T_{g,out}/T_{s,in}$ favours small channel sizes with the higher average emissivity, since

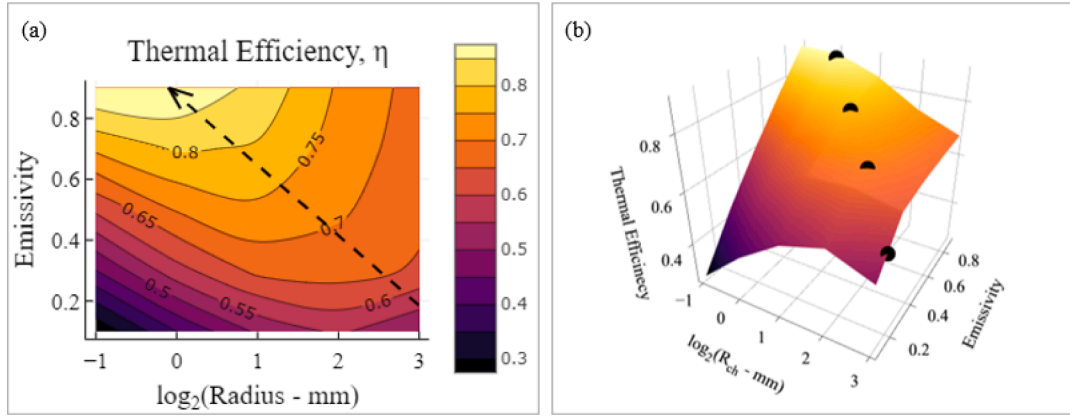


Fig. 19. Thermal efficiency plots for base case stage 1 overlaid with a) the (cut)line of designs that maximizes efficiency and b) the points (designs) selected to perform the comparison study.

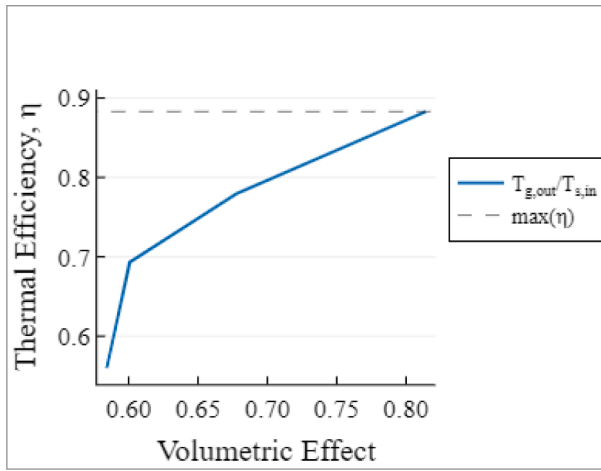


Fig. 20. Thermal efficiency vs. E_{vol} for selected base case receiver designs along the (cut)line (see Fig. 19) that maximizes efficiency.

those were the closest to achieving thermal equilibrium at the back. If simultaneous length optimization was carried out, it is likely that the ratio $T_{g,out}/T_{s,in}$ would have been maximized for a design that is close to maximizing thermal efficiency.

For definitions based on power ratios, $Q_{abs}/Q_{rad,loss}$ and $Q_{rad,loss}/Q_{BHS}$

end up being closest to their targets at the same design that minimizes radiative loss (low emissivity and a small channel size leading to view factors that quickly diminish with length). The temperature profile for this design, however, does not look qualitatively volumetric (see Fig. 23b) implying that these definitions are not valid. Additionally, Q_{BHS}/P_{ap} (or optical efficiency) is maximized for the highest emissivity (which is intuitive) and the largest channel size. The latter is probably because the single-channel assumption starts becoming invalid for channel sizes as large as 16[mm] and the flux received by a single channel at the centre of the receiver is much larger than that received by the surrounding channels. The temperature profile associated with this design is, also, very far from being qualitatively volumetric. Finally, Q_{abs}/Q_{BHS} leaned towards a region that minimized the total boundary heat source and, in turn, lowered thermal efficiency significantly compared to all other definitions. This leaves none of the definitions based on power ratios qualitatively valid, except for Q_{abs}/Q_{BHS} which severely reduces thermal efficiency.

Statistical-based evaluations of the local driving force (*RMSE*, *NMB*, and *NMSD*) all, approximately, fall on the contour for $\eta_{th} = 0.88$. *RMSE* and *NMB* (which are different forms of an averaged driving force) are minimized in regions of low porosity, where the gas and solid temperatures reach equilibrium, and the solid temperatures (and in turn driving forces) are lower (see Fig. 23c). *NMSD*, on the other hand, is minimized for the largest channel sizes, where the solid temperature exhibits a significant amount of variation. This large hot spot in the solid temperature profile, however, is not desirable in terms of ensuring solid stability or evenly utilizing the driving force.

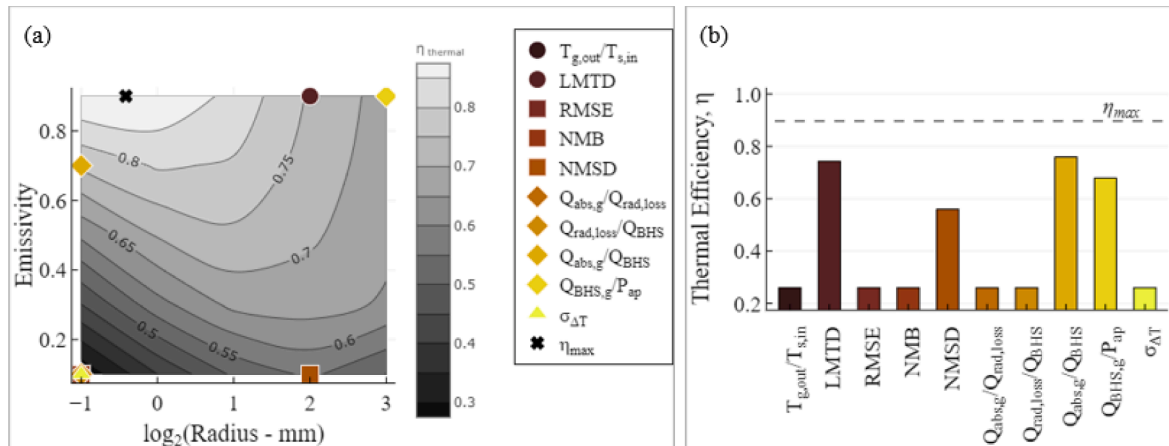


Fig. 21. a) Scattering designs favoured by each volumetric effect definitions over thermal efficiency contours for the base case receiver and b) comparing the efficiency of each of those designs.

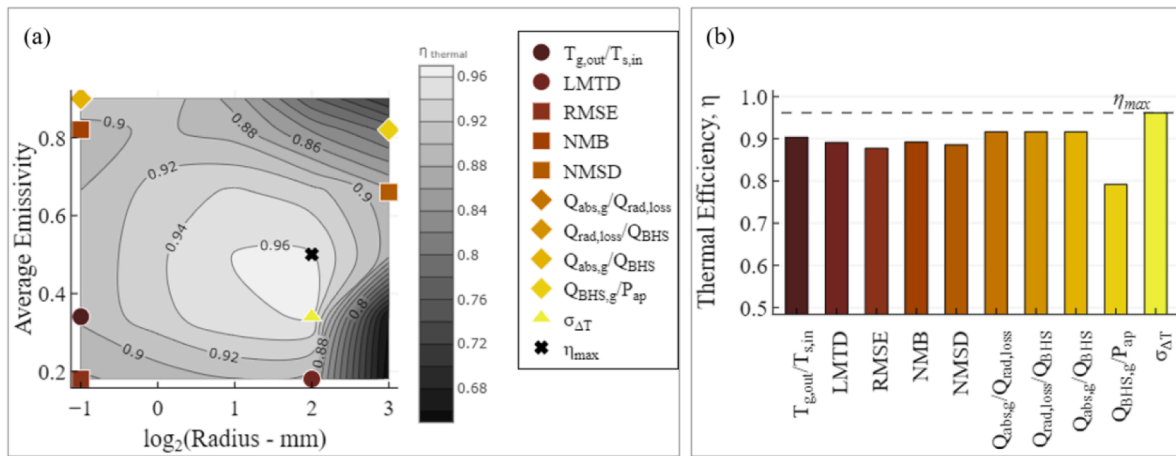


Fig. 22. a) Scattering designs favoured by each volumetric effect definitions over thermal efficiency contours for the specular case receiver and b) comparing the efficiency of each of those designs.

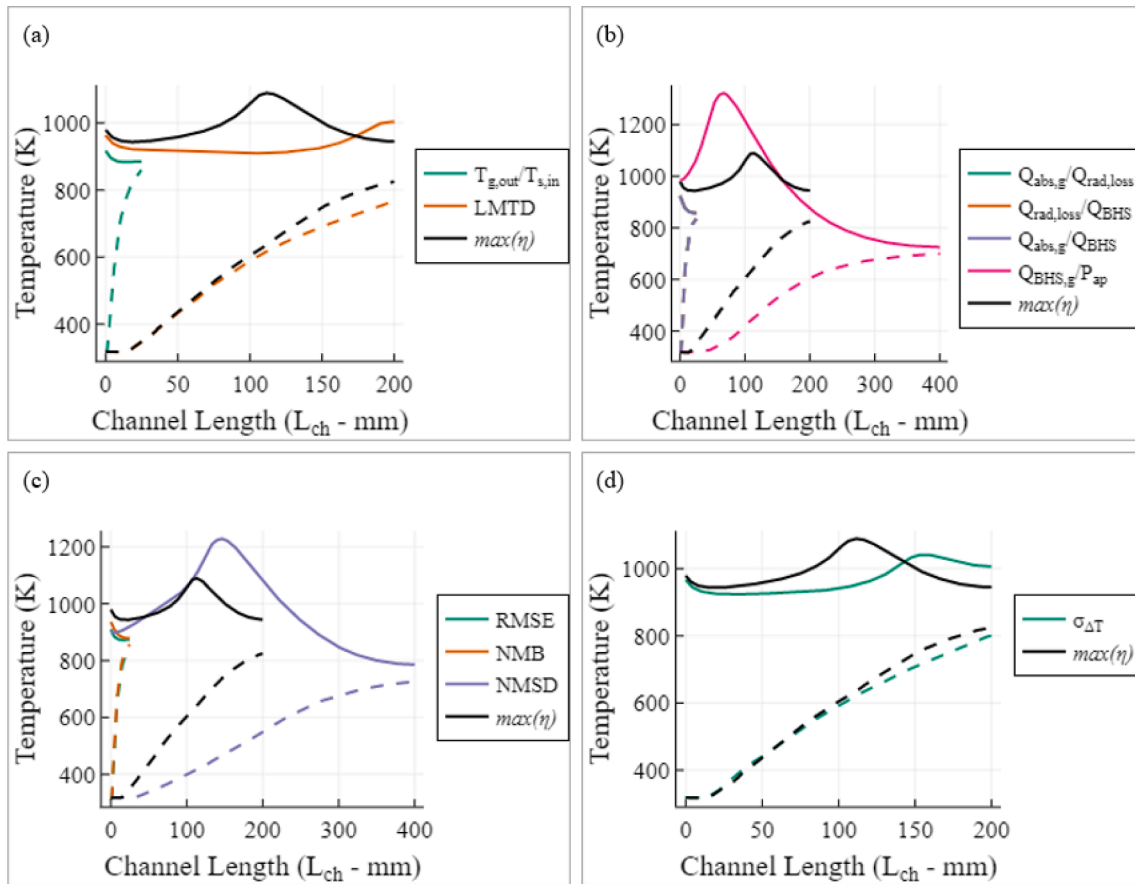


Fig. 23. Axial temperature profiles (solid lines for the solid, and dashed lines for the gas) at $P/\dot{m} = 700[\text{kJ/kg}]$ for specular receiver designs that maximize volumetric effect definitions based on: a) inlet and outlet temperatures, b) power ratios, c) statistical evaluations of the local driving force, and d) driving force non-uniformity.

Finally, the design that minimizes the driving force non-uniformity ($\sigma_{\Delta T}$) ends up being closest to the design that maximizes thermal efficiency. The temperature profile for that design, also, appears qualitatively volumetric (see Fig. 23d). However, extending the length of the receiver to achieve thermal equilibrium may affect the behaviour of this definition.

In summary, it appears that seeking the most evenly distributed driving force will arrive at the receiver design with the highest thermal

efficiency, but only when the receiver absorbs radiation volumetrically or evenly through its depth. Leading to the conclusion that the volumetric effect ought to be viewed as a combination of an evenly distributed heat source and an appropriate flow velocity (or $P/\dot{m}$ ratio) to extract that heat source evenly.

It is worth pointing out that unpublished work by Zaversky, *et al.* [43] claims the identification of a qualitatively 'volumetric' temperature profile using an indicator known as the Z-indicator – A ratio of the

boundary heat source to the heat absorbed by the gas in an inlet extinction region. That work did not show any examination of BHS profiles, but their experimental observations do agree with our conclusions. They observed that for a collection of conventional uniform porosity foam receivers, many designs will not exhibit their highest thermal efficiency when volumetric effect (described using the Z-indicator) is maximized. These results can be contextualized within the current discussion as a conventional foam receiver will absorb most primary radiation in a shallow frontal region and only exhibit a volumetric temperature profile at higher incident fluxes (or lower $P_{ap}/\dot{m}$ ratio) as a result of increased radiative cooling of the front surface due to its high frontal solid temperatures and view factor of nearly unity. The Z-indicator developed by Zaversky, *et al.* [43] supports this explanation as it exhibits a value of less than 1 (no volumetric effect) for all cases where the gas exit temperature exceeds 1100 °C, even though the solid temperature profile does appear to be volumetric and increases monotonically from inlet to outlet.

5. Conclusions

In this work, varied reflectivity was explored as a route to attaining true volumetric behavior. Varied-reflectivity receivers have the opportunity to maintain the simple robust design of commercially proven honeycomb receivers, while achieving similar performance to the successful, although complex, varied-porosity receivers reported in literature. The model used to explore this concept was developed using a detailed approach for the treatment of primary radiation and convective modelling and verified using simulation data published in literature. Though many papers publish their results as a source of validation data, it is often not usable because insufficient data about the HFSS or the primary radiation source is provided.

The verified model was used to explore the varied-reflectivity receiver concept starting with a number of diffuse reflectivity distributions. Those showed slightly improved thermal performance, but due to increased frontal absorption and not improved radiation penetration or volumetric utilization. As a second attempt, a specular reflectivity distribution was attempted, and the thermal performance results were on-par with other successful varied-porosity designs in the literature. A maximum thermal efficiency of 96% was obtained and a qualitatively more volumetric temperature profile was achieved. The optical properties of this specular receiver were assumed, but are very similar to those of MAX-phase Cr_2AlC , which has been shown to be stable under concentrated solar irradiance. Though the specular receiver showed a more uniform heat source distribution along its depth, currently available quantitative definitions of the volumetric effect (mainly $T_{g,out}/T_{s,in}$) could not detect this behavior.

Therefore, an investigation on quantitative definitions for the volumetric effect was carried out, motivated by the results of the specular and diffuse reflectivity receivers. Different definitions for the volumetric effect were screened; first, for one receiver under different operating conditions, then, for several receiver designs under the same operating conditions. The latter approach yielded more elucidating results. When the screening was applied to the base case (uniform reflectivity) receiver as well as the specular case receiver to see how the designs targeted by each definition change between receivers that do not exhibit the volumetric effect (the base case) to receivers that do exhibit it (the specular case). Of the 10 tested definitions, the ones that showed the best agreement with qualitatively volumetric temperature profiles were $\sigma_{\Delta T}$ and $T_{g,out}/T_{s,in}$, but only when the receiver exhibits volumetric absorption (i.e. the specular case). Experimental literature results, also, supports our conclusion that thermal efficiency and volumetric behaviour only target the same design when the receiver absorption (or BHS profile) is truly volumetric and the incident irradiance and $P/\dot{m}$ ratio allow complete extraction of the absorbed irradiance.

Nevertheless, the reader is encouraged to explore other volumetric

effect definitions as well as other approaches for screening them by accessing the raw data (temperature profiles and global power evaluations for all the presented receiver cases) used in this work through the repository [\[https://github.com/tamu-edu/tamuq-chen-secarelab-receiver\]](https://github.com/tamu-edu/tamuq-chen-secarelab-receiver).

CRedit authorship contribution statement

Fathya Salih: Conceptualization, Methodology, Software, Data curation, Writing – original draft, Visualization. **Konstantinos E. Kakosimos:** Conceptualization, Methodology, Writing – review & editing, Supervision, Funding acquisition, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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